

Annotated Table of Contents Coursework COMP1649

HumanComputer Interaction and Design (2024-2025)

Below is an example of how a table of contents for your coursework report COULD look like. You can use a different structure to what is shown here as long as the relevant content is covered.

Your report should demonstrate sufficient English proficiency, (critical and concise) writing, structuring and report formatting skills. In-text citations and references need to be correctly formatted in Harvard style. The word count for the report should be about 2000-3000 words long. You need to include at least 8 academic in-text citations/references (e.g. academic conference papers, journal papers, chapters from academic books etc.) to support your writing.

The format of the prototype must be an Axure file (.rp) unless agreed with the lecturer otherwise.

1 Introduction

This report details the development of Azuden, a mobile application created to enhance the riding experience of motorcyclists by playing music while tracking their routes. The design process incorporated human-computer interaction (HCI) principles, extensive user research, wireframing, and prototyping to ensure the application was aligned with real-world user needs. The document outlines the application's goals, research findings, design decisions, and challenges during development, providing insight into how Azuden progressed from prototype to completion.

2 Background literature

2.1 HCI Research

During the development of the Azuden application, human-computer interaction (HCI) research has provided key insights for designing and integrating functions to enhance user experiences. Research by Kingsley et al. (2020) discusses the application of augmented reality (AR) for delivering real-time alerts and contextual information, thereby improving safety and situational awareness for users during movement. Similarly, Imsirovic and Kumnova (2019) highlight how music integration within HCI applications can enhance user engagement and boost motivation, making the experience more enjoyable and immersive. In addition, Leung et al. (2007) evaluate haptically augmented touchscreen graphical user interface (GUI) elements under cognitive load, demonstrating that intuitive touch controls, including haptic feedback, can reduce user cognitive strain and enhance interaction in dynamic environments. Finally, Greasley and Lamont (2011) emphasize the importance of visual cues and their role in guiding user interactions effectively, helping to minimize distractions and maintain user focus on primary tasks.

From these findings, several key requirements for Azuden can be established. The application should include a simple, user-friendly interface with large buttons for easy function switching, integrate music seamlessly to enhance user engagement and incorporate clear visual cues to assist users without increasing cognitive load. These elements collectively aim to provide motorcyclists with a safe, intuitive, and engaging experience, combining route tracking and music playback effectively.

2.2 HCI Theory

2.2.1 Cognitive Psychology

Research by (Norman, 2013) emphasizes the importance of cognitive load in interface design, suggesting that interfaces should minimize the amount of information users must retain while interacting with an application. This is particularly relevant for Azuden as it aims to balance route tracking and music playback, ensuring users can focus on the road without being overwhelmed by complex inputs. Cognitive processes such as attention and perception are vital; Azuden's interface needs to prioritize essential functions with clear labels and streamlined actions, reducing cognitive overload (Norman, 2013). This approach supports intuitive use and helps users remain engaged while maintaining safety.

2.2.2 Interaction Design Theory

According to (Sharp et al., 2019), interaction design focuses on creating meaningful experiences between users and digital systems. For Azuden, a conceptual model incorporating recognizable metaphors—such as music player controls and navigation icons—enhances user familiarity and ease of navigation. Prototyping at different fidelities (low and mid-fidelity) enables iterative testing and refinement, aligning the application's functionality with user expectations and interaction preferences (Sharp et al., 2019). This iterative process ensures the product's design aligns with practical user needs and expectations.

2.2.3 Modes of Interaction

According to (Norman, 2013), touch-based interactions are among the most intuitive modes for mobile devices. Azuden leverages touch interfaces, enabling users to swipe, tap, and interact with large buttons for music controls and route selection. Touch gestures facilitate hands-free navigation, enhancing safety and user control during rides. This mode is essential in the context of motorcycling, where simple and responsive interactions are crucial for minimizing distractions (Norman, 2013).

2.2.4 Types of Interaction

Interaction types play a critical role in shaping how users engage with an application. As outlined by (Cooper et al., 2014), instructing and manipulating interactions are especially relevant to Azuden. Users can instruct the app to play music, start navigation, or manage settings through touch commands, while manipulation involves engaging with interactive elements on the screen. This approach ensures that users can achieve their tasks intuitively and efficiently, aligning with the application's goals of safety and engagement during travel.

2.2.5 Design Principles

Design principles such as visibility, consistency, and feedback form the foundation for effective interaction design (Nielsen, 1994). Azuden's interface emphasizes visibility by displaying essential controls prominently. Consistency in layout and interaction patterns helps users predict system responses, while real-time feedback—such as music playback status and route progress—ensures users understand their current state. These principles collectively enhance user trust, usability, and overall satisfaction.

2.2.6 Design Pattern

As discussed by (Tidwell, 2010), design patterns provide reusable solutions to common interaction challenges. Azuden employs design patterns like a "music player control panel" for seamless music management and "navigation breadcrumb" trails for route tracking. These patterns enhance user familiarity and reduce learning curves, enabling motorcyclists to interact safely and effectively with the application. By aligning with established design patterns, Azuden ensures a user-centered and consistent experience.

3 Product Design

This app for motorcyclists combines GPS-based ride tracking with music streaming. Users can select routes, review trip history, and enjoy music through customizable playlists, favorites, and simple song switching. Account management ensures secure handling of personal information, while settings provide feedback options and privacy policy details. It emphasizes a safe, enjoyable.

3.1 Conceptual Design

3.1.1 User Requirements

The Azuden application aims to enhance the motorcyclist experience by seamlessly integrating route tracking and music playback. Key user requirements include a distraction-minimized interface for safe travel, easy access to core features like GPS navigation and music with minimal interactions, and responsive controls with clear notifications for smooth transitions. Users also need customizable options for routes, music playlists, and app settings, while ensuring data security and privacy for account management and travel history. These requirements ensure a user-friendly, engaging, and safe experience tailored to motorcyclists.

3.1.2 Problems and Solutions

The primary problem was developing a mobile application that allows users to track their travel routes and listen to music while ensuring safety, minimizing distractions, and maintaining performance during GPS and music usage. To address user distraction, a simplified interface with large buttons and intuitive navigation was implemented, allowing for easy and safe interactions while on the move. System optimization ensured seamless transitions and efficient operation of GPS tracking alongside music playback, enhancing overall user experience and safety.

3.1.3 Hierarchical Model

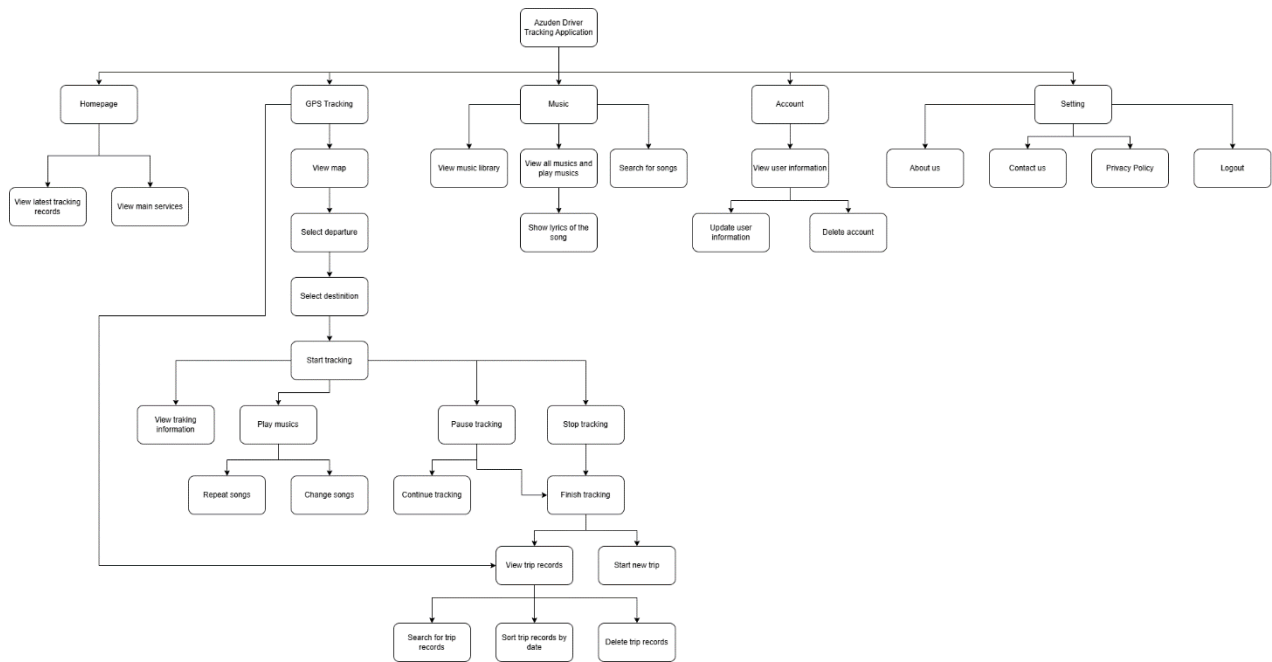


Figure 1. Hierarchical Model

Figure 1 illustrates the hierarchical model of the Azuden Driver Tracking Application, showcasing the structure of key features and interactions within the app. The model starts from the homepage, branching into major sections such as GPS Tracking, Music, Account, and Settings. Each section contains subfunctions, for instance, GPS Tracking includes options like viewing maps, selecting departure and destination, and managing trip tracking. Music functions offer viewing and playing music, while Account allows user information management. The hierarchical structure emphasizes streamlined navigation, offering users a clear, organized flow for accessing features, ensuring minimal distraction and intuitive usability while driving.

3.1.4 Conceptual Model

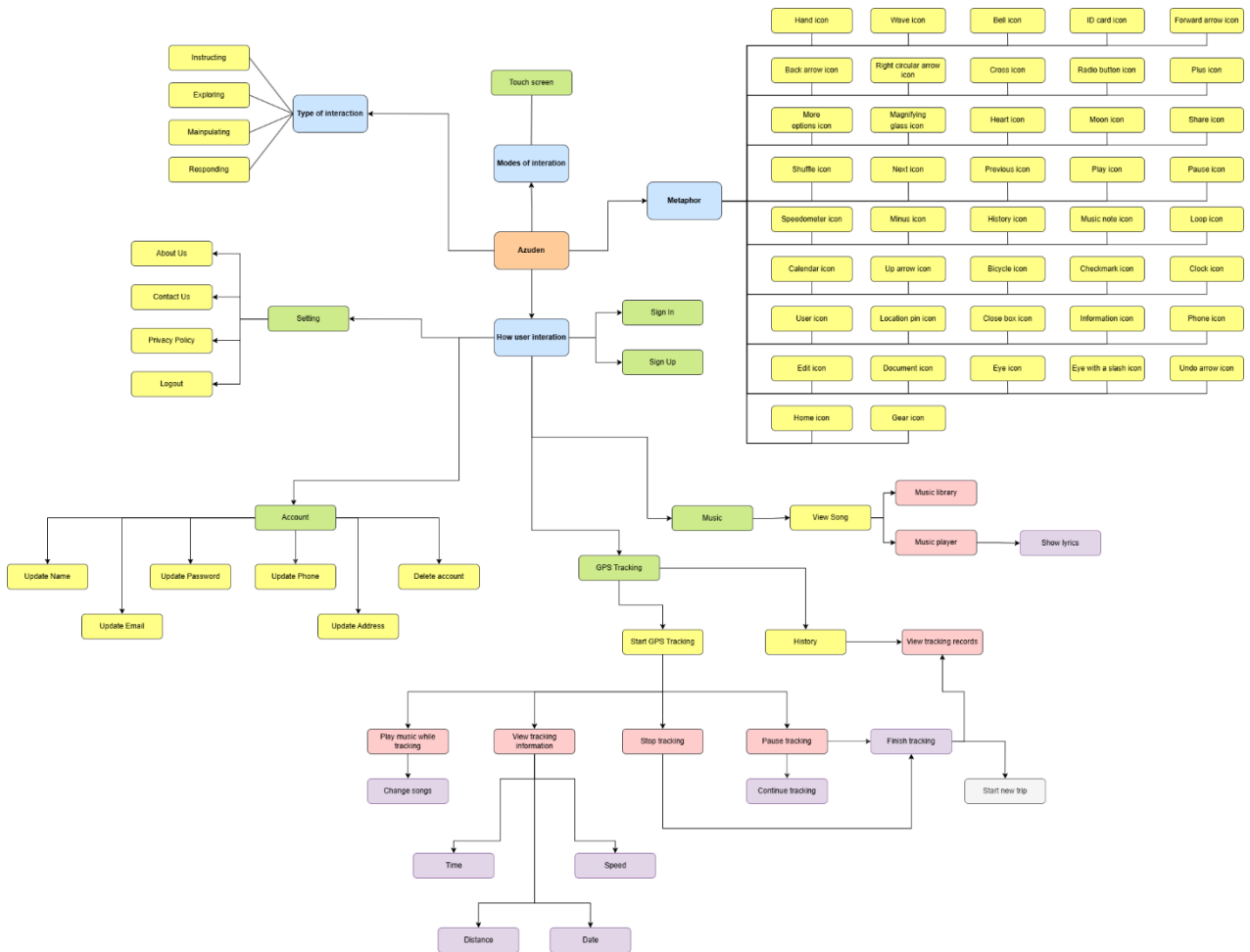


Figure 2. Flowchart of the Conceptual Model for the Azuden, showcasing user interactions, interaction types, and metaphors.

Figure 18 is a conceptual model of the Azuden application, showcasing its core components, including interaction types, modes of interaction, user interactions (such as settings, account management, and GPS tracking), and metaphors represented by various icons. The model highlights how users interact with features like music playback, trip tracking, and account updates, emphasizing seamless integration of music and GPS functions for a streamlined user experience.

3.1.5 Metaphors

The Metaphors section in Table 1 highlights icons used in the Azuden App to visually represent features and actions. These icons help users quickly understand and navigate functions, enhancing app intuitiveness and ease of use.

Table 1. Metaphors table

Icon	Description
	Hand icon - represents interaction, such as tapping the screen
	Wave icon - used for greetings or welcoming animations

	Bell icon - indicates notifications or alerts
	ID card icon - represents user identification or profile details
	Forward arrow icon - access or move forward in navigation
	Back arrow icon - navigate back to the previous screen
	Right circular arrow icon - navigate to the page the icon belongs to.
	Cross icon - cancel or close the current action
	Radio button icon - select a location to start moving
	Plus icon - add a new item or entry
	More options icon (three dots) - access additional menu options
	Magnifying glass icon - search within the app or list
	Heart icon - mark as favorite or like
	Moon icon - toggle night mode or dark theme
	Share icon - share content with others
	Shuffle icon - enable random playback order
	Next icon - skip to the next item
	Previous icon - return to the previous item
	Play icon - start playback
	Pause icon - pause playback

	Speedometer icon - displays speed or performance metrics
	Minus icon - remove or reduce an item or value
	History icon - access to travel history page
	Music note icon - access or control music features
	Loop icon - repeat content or function continuously
	Calendar icon - schedule or view dates and events
	Up arrow icon - scroll or move upward
	Bicycle icon - display speed and vehicles moving on the route
	Checkmark icon - confirm completion or selection
	Clock icon - display or manage time-related features
	User icon - view or manage user profile
	Location pin icon - mark or identify a location
	Close box icon – delete or hide an item
	Information icon - display detailed information or help
	Phone icon - make a call or access contact options
	Edit icon - modify or update content
	Document icon - open or view documents and records
	Eye icon - view content or make visible

	Eye with a slash icon - hide content or make invisible
	Undo arrow icon - logout
	Home icon - navigate to the home screen
	Gear icon - access settings or configuration options

3.2 Design Principles

The system is designed according to the user interface design principles proposed by Jakob Nielsen and Don Norman.

3.2.1 Jakob Nielsen's Principle

- Visibility of system status



Figure 3. Display of key trip metrics to maintain user awareness of system status.

Figure 18 shows Nielsen's principle of visibility of system status in action. Key trip information such as distance (KM), average speed (KM/H), date, and travel duration is displayed. This ensures users are consistently aware of the current system status and trip progress, providing immediate and understandable feedback throughout their journey.

- Match between the system and the real world

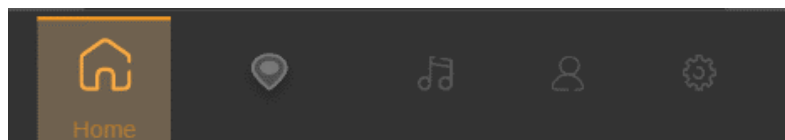


Figure 4. Navigation bar with intuitive icons matching real-world concepts for user-friendly interaction.

Figure 19 shows a navigation bar that illustrates Nielsen's match between system and the real world. The icons used for navigation, such as Home, Location, Music, User, and Settings, employ universally recognized symbols and labels that align with users' real-world experiences and expectations. This intuitive design ensures users can easily understand and interact with the navigation bar, making their journey through the app seamless and predictable.

- User control and freedom



Figure 5. User control elements for navigation and playback, enhancing flexibility and freedom.

Figure 20 illustrates Nielsen's user control and freedom. The first image shows music playback controls, including options for shuffle, skip, pause/play, and other features, enabling users to navigate and control their music experience with flexibility. The second image features "Back" and "Let's go" buttons, offering users the freedom to either proceed or return, ensuring they have control over their navigation choices and actions within the app.

- Consistency and standards

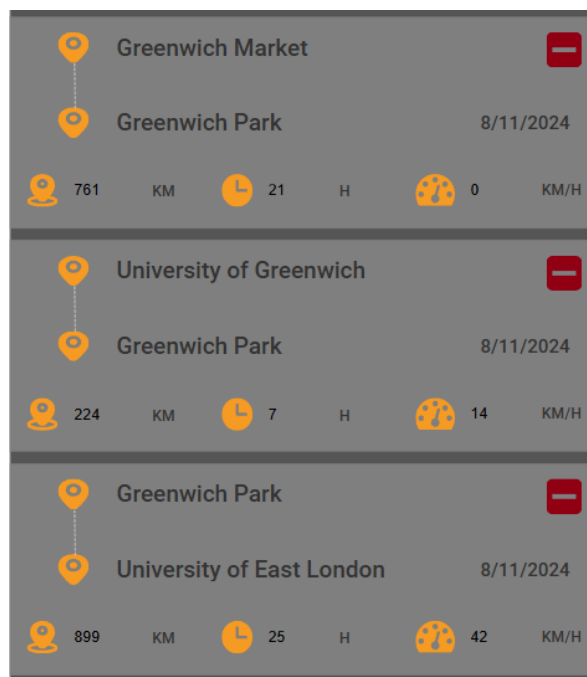


Figure 6. Consistent design and formatting for trip history entries, ensuring predictability and ease of use.

Figure 21 demonstrates Nielsen's consistency and standards. The trip history entries maintain a consistent design pattern with uniform formatting, icons, and data structure across different trips. This consistent use of layout and visual elements ensures that users can quickly recognize and understand each entry, reducing cognitive load and providing a predictable experience when viewing their travel history.

- Error prevention

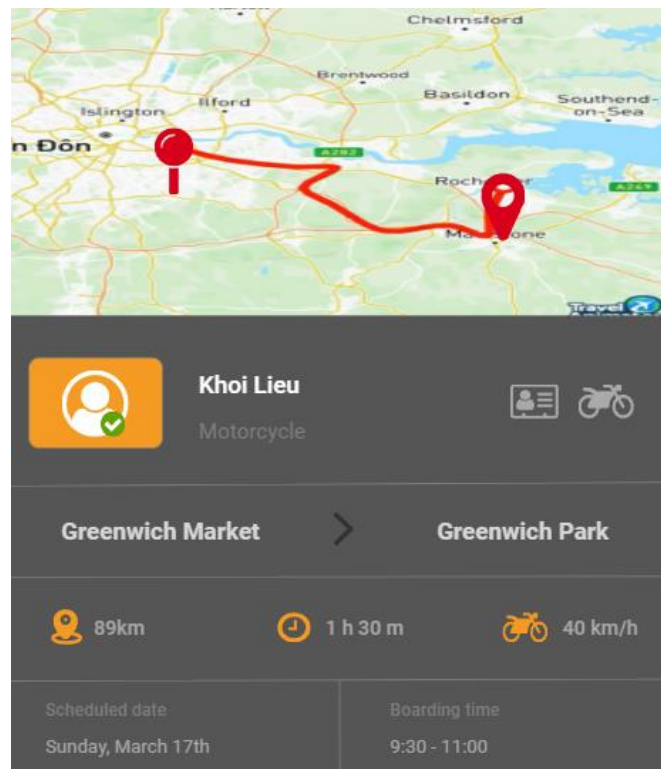


Figure 7. Trip confirmation screen to help users review and prevent potential errors before starting.

Figure 22 illustrates Nielsen's error prevention. This confirmation screen allows users to review essential trip details, including the route, distance, estimated travel time, speed, and scheduled times before starting their journey. By providing a summary for verification, the app minimizes the chance of users selecting incorrect information or proceeding without reviewing critical trip details, effectively preventing potential errors in their journey setup.

- Recognition rather than recall

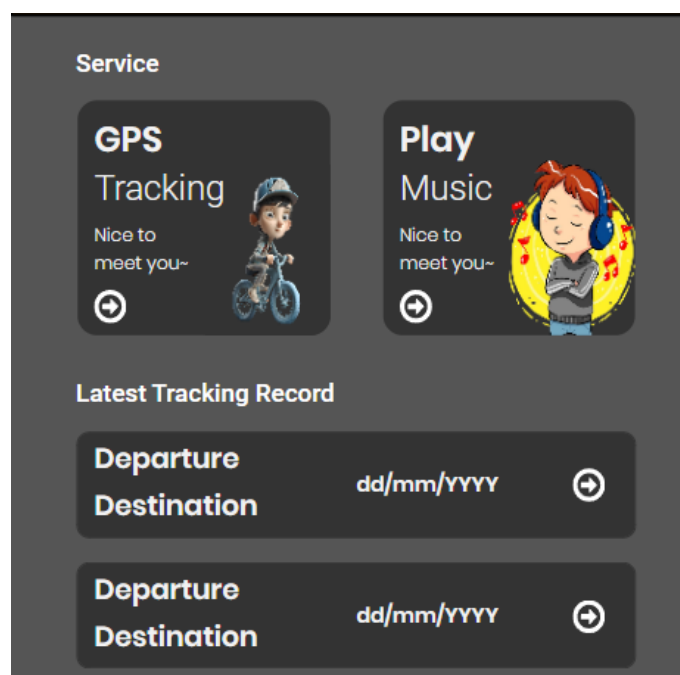


Figure 8. Home screen with clearly labeled main functions, reducing memory load and enhancing recognition.

Figure 23 illustrates Nielsen's recognition rather than recall. The home screen presents clearly labeled and visually distinct options for GPS tracking and music playback, as well as a section for recent trip records. This design minimizes the need for users to remember specific actions or navigation paths, allowing them to recognize and easily access core functions of the app with minimal effort.

- Flexibility and efficiency of use

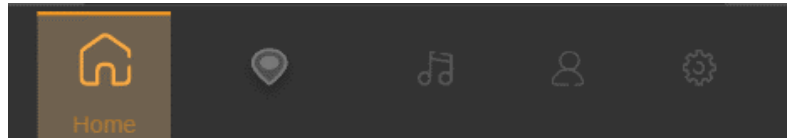


Figure 9. Navigation bar for flexible and efficient access to app sections, suitable for all user levels.

Figure 24 demonstrates Nielsen's flexibility and efficiency of use. The navigation bar provides intuitive icons that allow users, including new and experienced ones, to easily access and switch between different app sections.

- Aesthetic and minimalist design

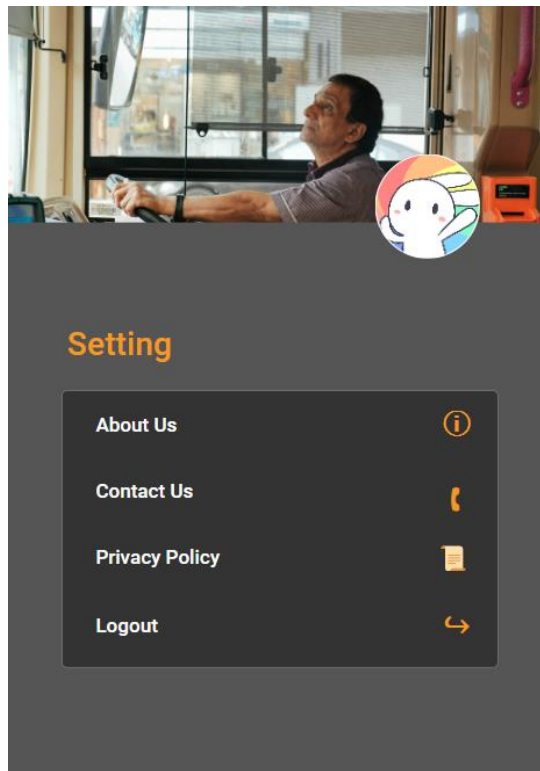


Figure 10. Minimalist settings screen with essential options for a clean, user-friendly interface.

Figure 25 illustrates Nielsen's aesthetic and minimalist design. The settings screen features a clean, uncluttered layout with a minimalist design, displaying only essential options such as "About Us," "Contact Us," "Privacy Policy," and "Logout." The use of icons and simple labels provides a visually pleasing interface.

- Help users recognize, diagnose, and recover from errors

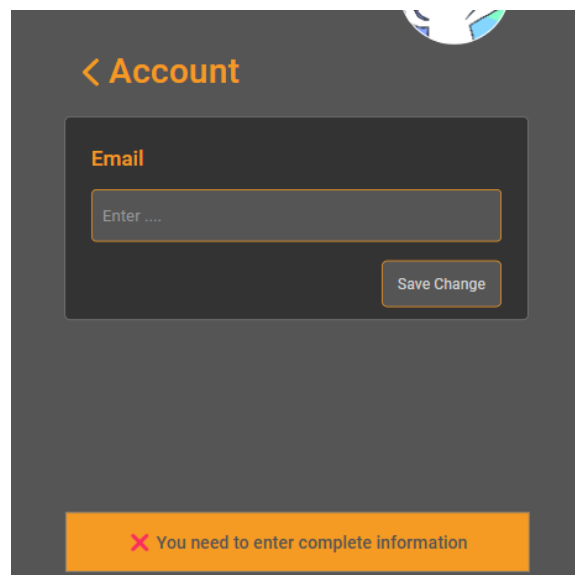


Figure 11. Clear error message guiding users to correct incomplete information during form submission.

Figure 26 illustrates Nielsen's help users recognize, diagnose, and recover from errors. When a user attempts to save changes without entering complete information, the system provides a clear error

message at the bottom of the screen, indicating the issue and guiding the user to take corrective action.

- **Help and documentation**

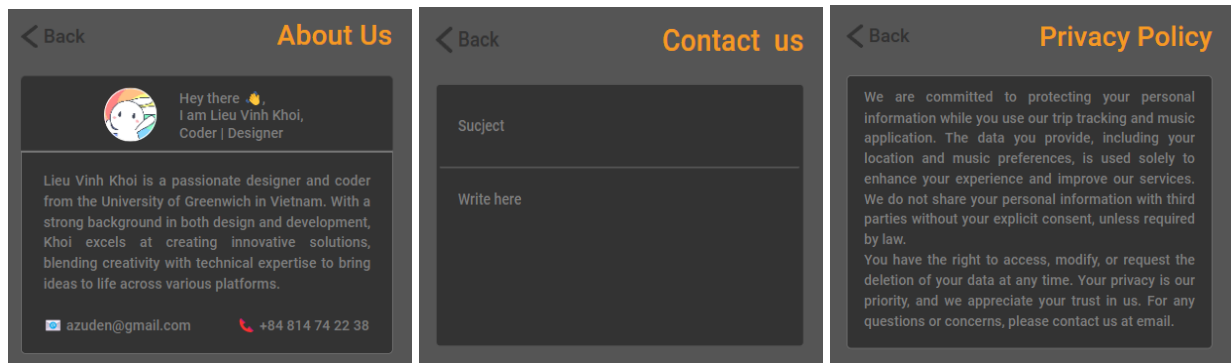


Figure 12. Sections for user support and essential app documentation, ensuring easy access to information and help.

Figure 27 demonstrates Nielsen's help and documentation. The "About Us," "Contact Us," and "Privacy Policy" sections provide essential information to assist users in understanding the app's purpose, reaching out for support, and knowing their data privacy rights. These elements ensure users have access to relevant documentation and support options when needed.

3.2.2 Don Norman's Principle

- **Visibility**

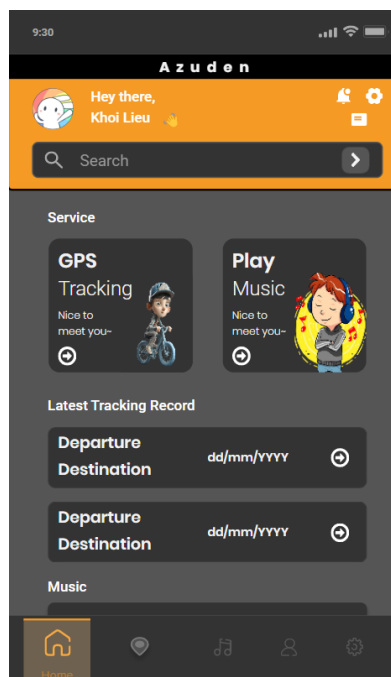


Figure 13. The functions, buttons, and icons are displayed more clearly on the Home page.

Figure 28 shows that functions, buttons, and icons in the Azuden application are always carefully placed in appropriate and clear positions, helping users to easily operate and choose immediately.

- **Feedback**

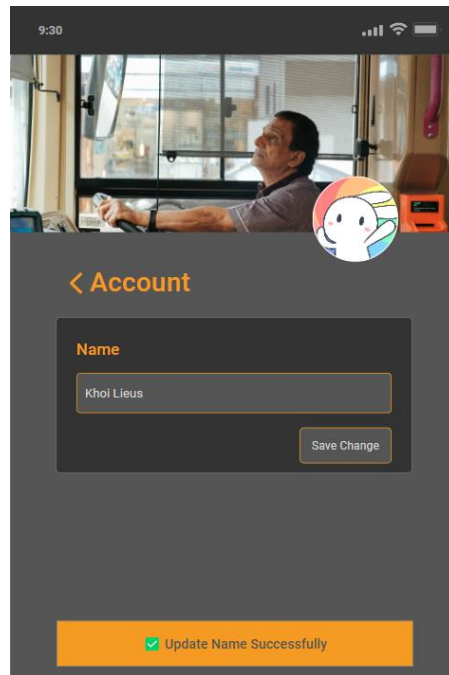


Figure 14. Feedback provided through a success message after updating user information.

Figure 29 illustrates Don Norman's feedback principle. Upon updating their name, users receive immediate confirmation with a "Update Name Successfully" message, ensuring clarity and confidence in their action.

- Affordance

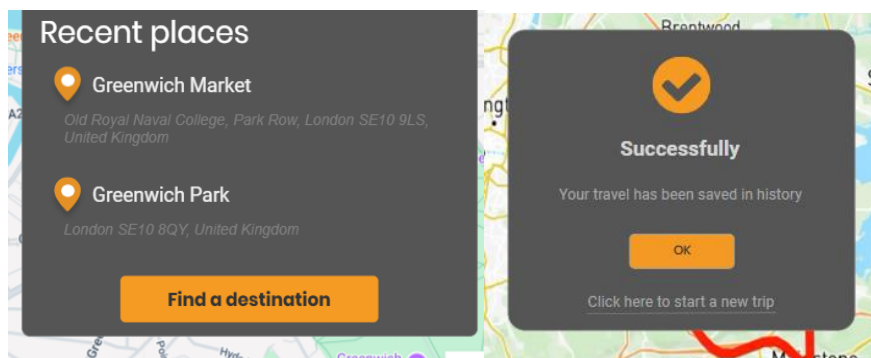


Figure 15. Prominent buttons and icons in the app

Figure 30 shows that the buttons, and icons in the Azuden application are all designed with prominent and suitable colors, the hints or signs in this design show that users can easily recognize to use them.

- Mapping

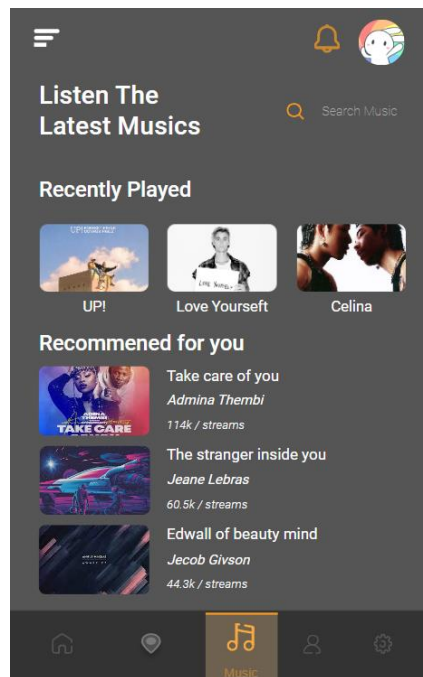


Figure 16. Prominently display music icon in navigation bar with navigation to music page.

Figure 31 illustrates Don Norman's mapping principle. The highlighted music icon in the navigation bar while the displayed music screen, provides a clear and logical connection between the control (music icon) and its result (navigation to the music page). This mapping ensures users can easily understand and predict the outcome of their navigation actions.

- Constraints

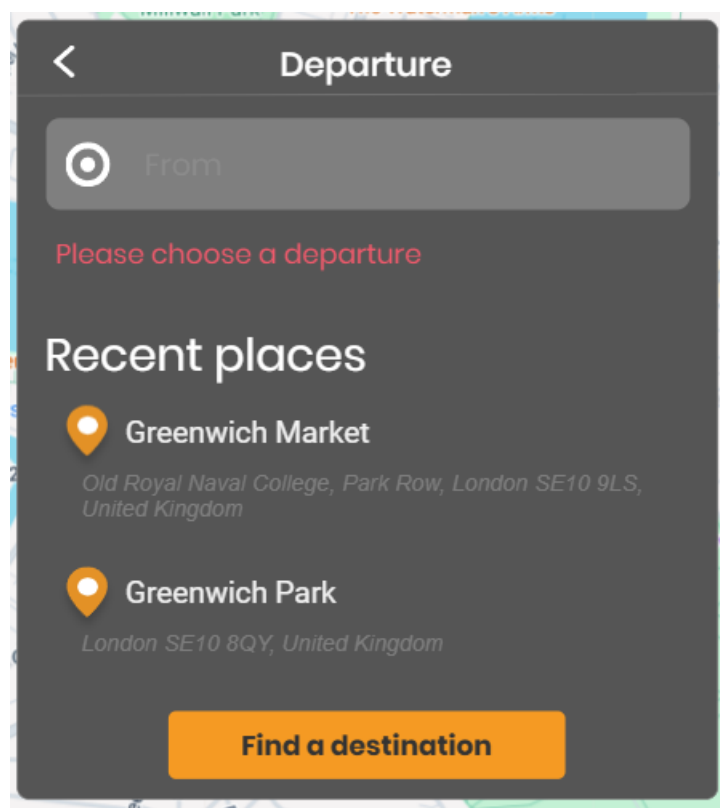


Figure 17. Constraint applied to ensure users select a departure point before proceeding.

Figure 32 demonstrates Don Norman's constraints principle. When users attempt to proceed to the

destination selection without choosing a departure point, the system restricts this action and displays a prompt, "Please choose a departure." This constraint ensures users follow the correct process and prevents potential errors by guiding them through required steps.

- Consistency

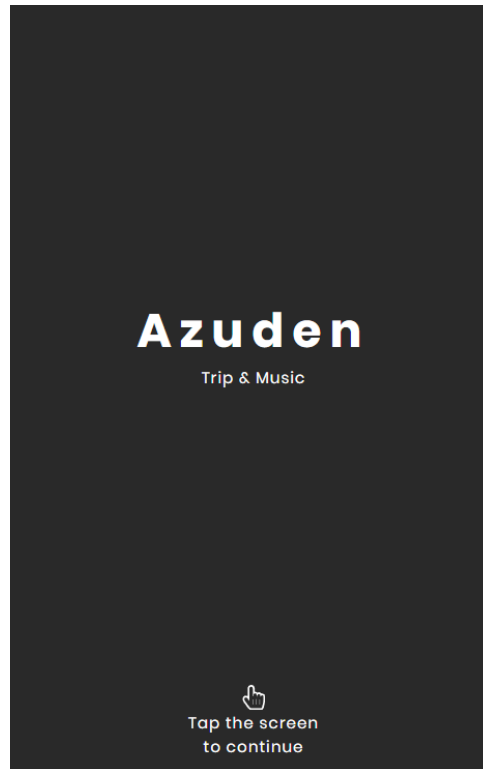


Figure 18. Consistent welcome screen design with clear instructions for user interaction.

Figure 32 illustrates Don Norman's consistency principle. The welcome screen maintains a consistent design pattern by displaying a clear instruction, "Tap the screen to continue," which aligns with the app's user flow. This consistency helps users easily understand the interaction needed and ensures a predictable navigation experience throughout the app.

4 Prototype

4.1 Low-fidelity Prototype

- Mobile Screen

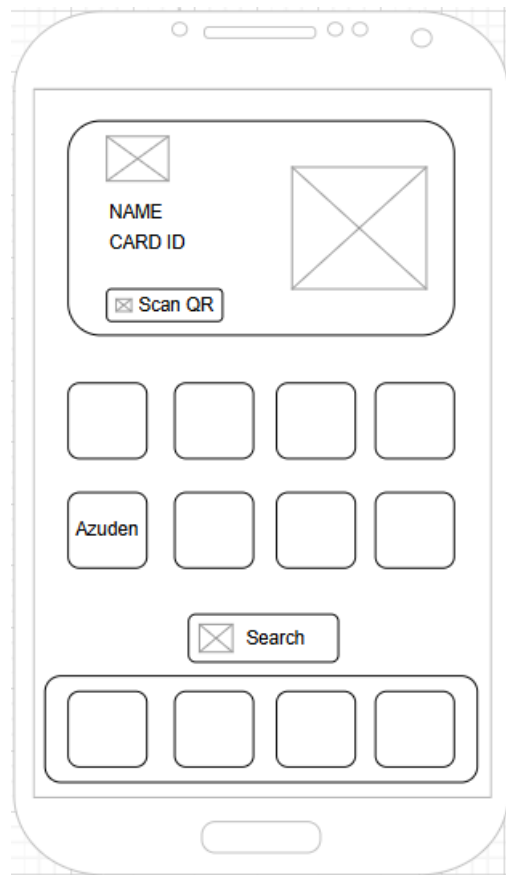


Figure 19. Mobile Screen WireFrame

- Welcome Screen

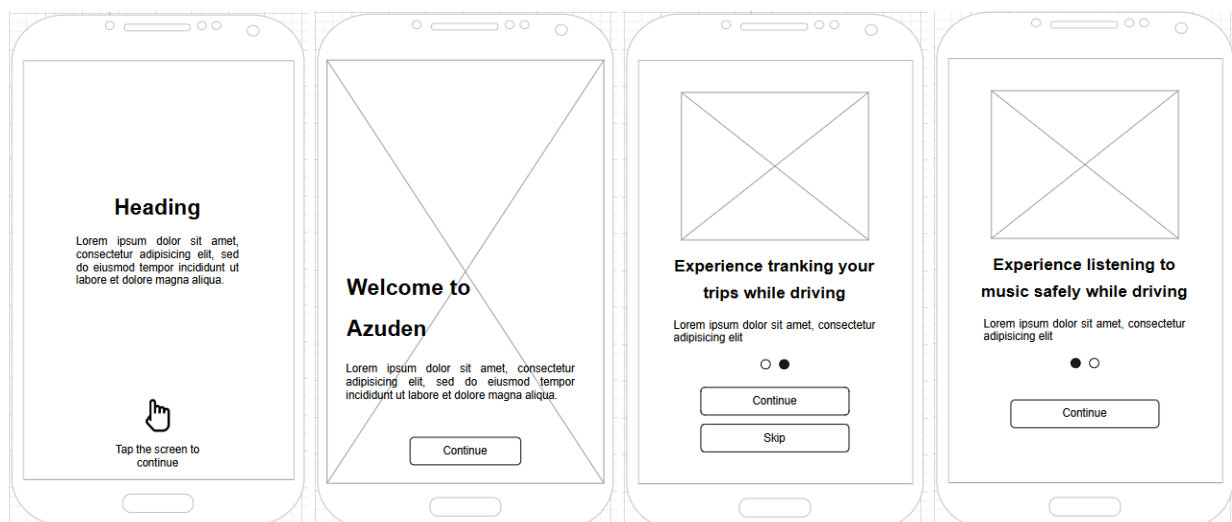


Figure 20. Welcome Screen WireFrame

- Sign In and Sign Up Screen

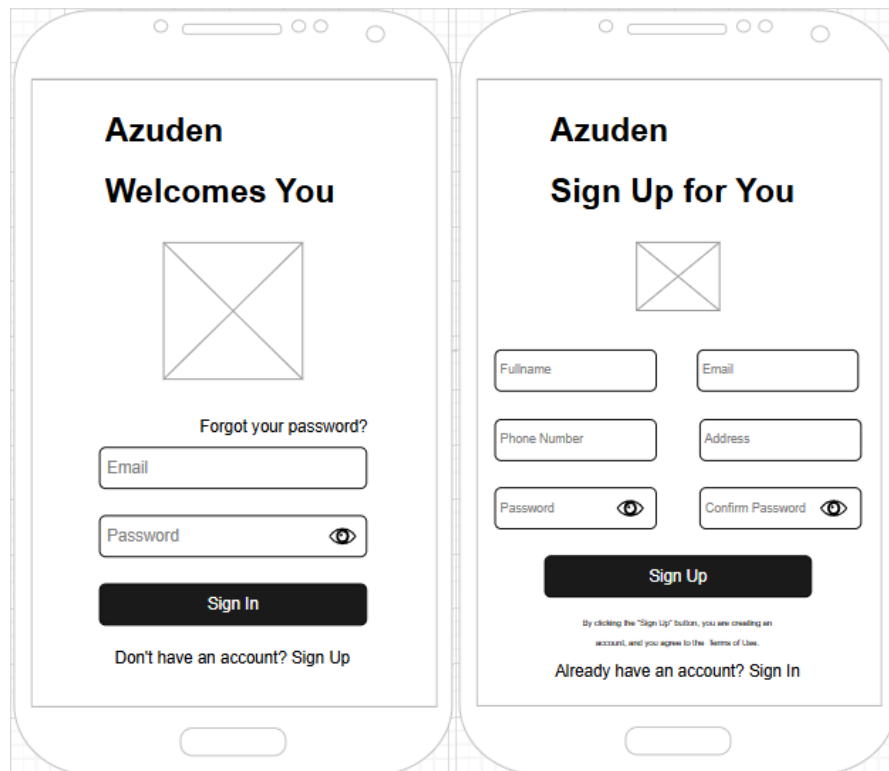


Figure 21. Sign In and Sign Up Screen WireFrame

- Home Screen

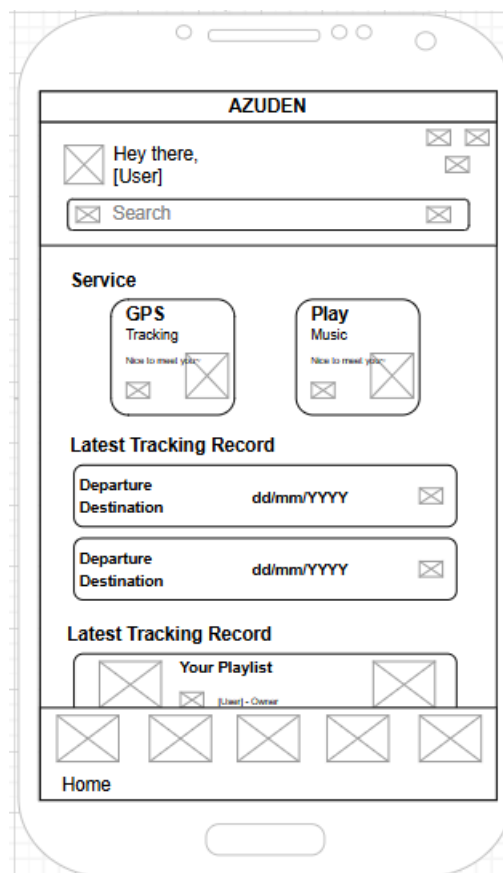


Figure 22. Home Screen WireFrame

- Route Selection Screen

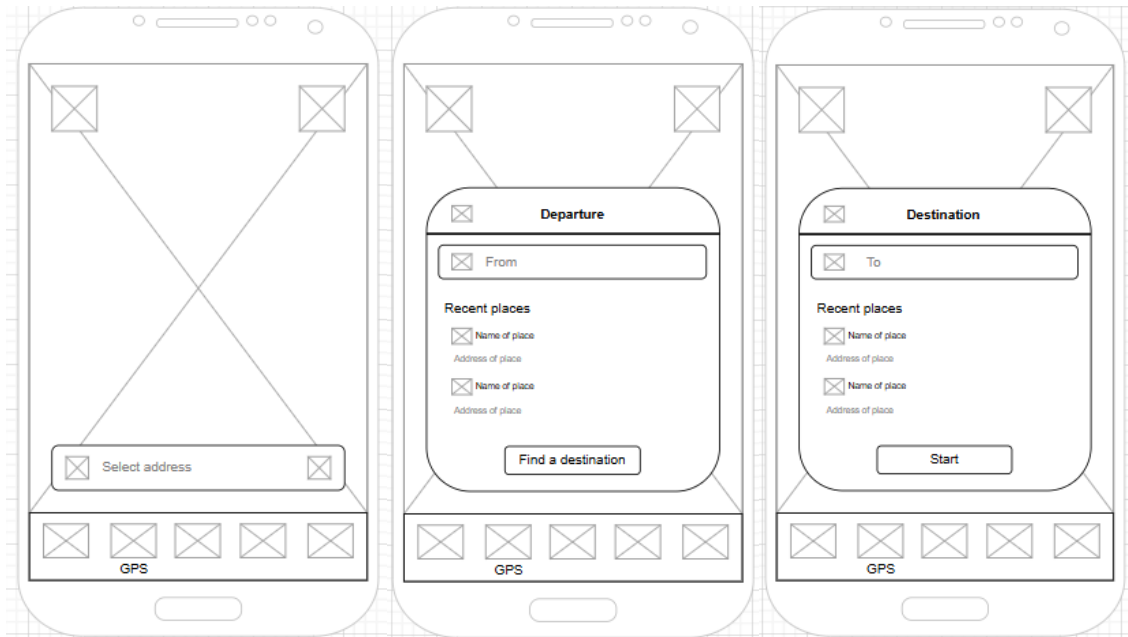


Figure 23. Route Selection Screen Wireframe

- Route Start Confirmation Screen

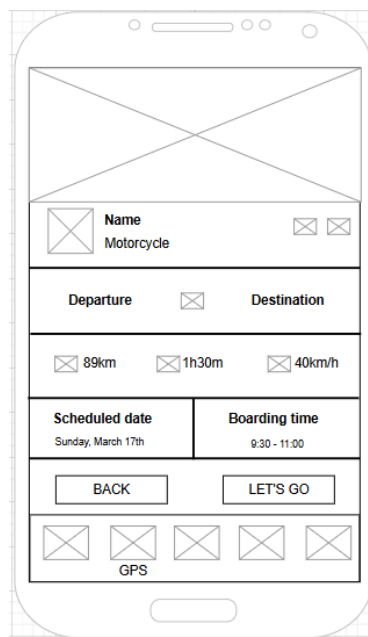


Figure 24. Route Start Confirmation Screen Wireframe

- Travel Route Tracking Screen



Figure 25. Travel Route Tracking Screen Wireframe

- Music Screen While Tracking The Route

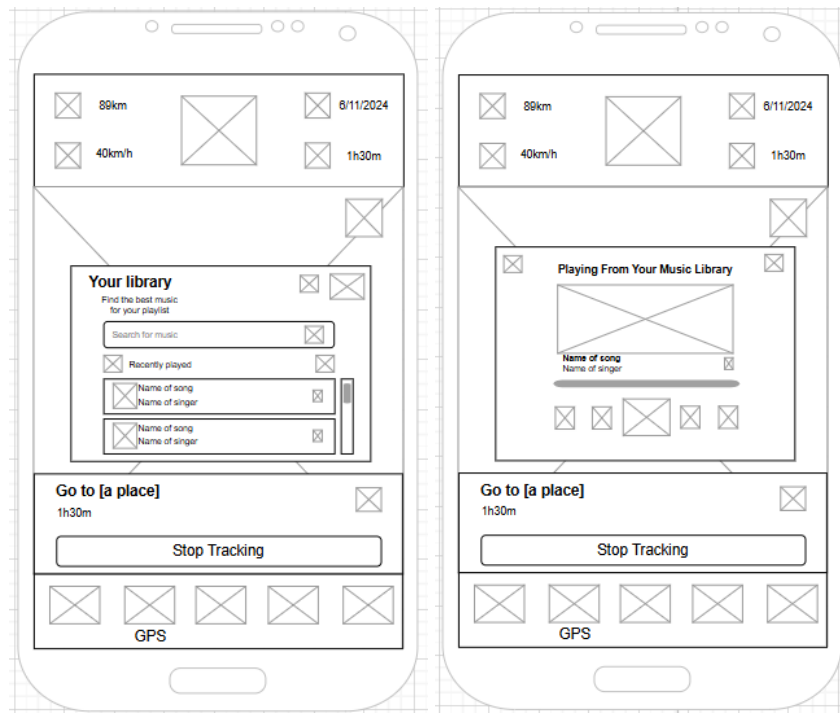


Figure 26. Music Screen While Tracking The Route

- Travel Route Completion Screen

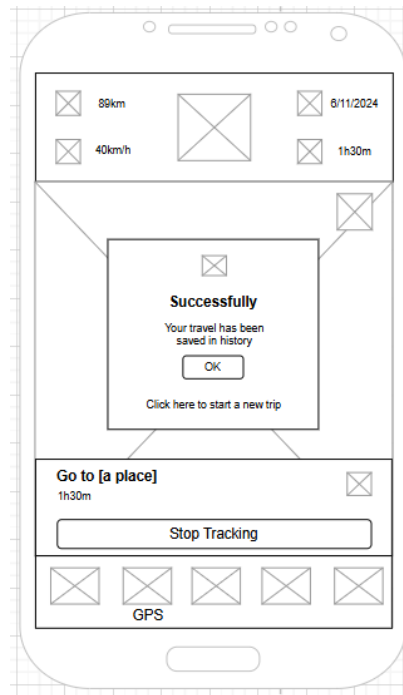


Figure 27. Travel Route Completion Screen Wireframe

- Tracking Record History Screen

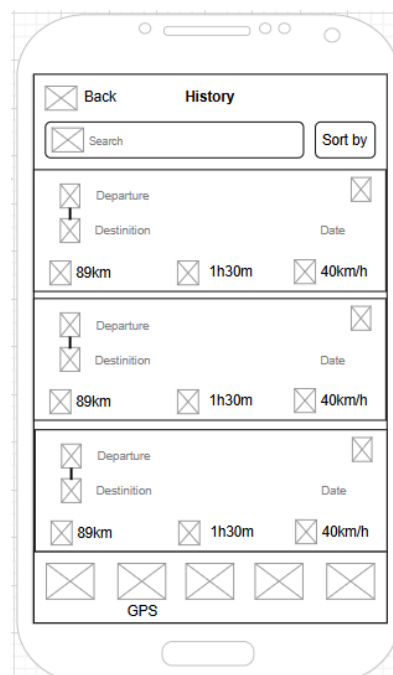


Figure 28. Tracking Record History Screen Wireframe

- Music Manager Screen Wireframe

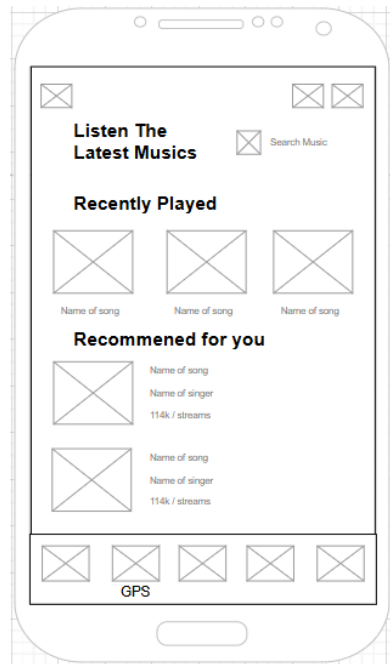


Figure 29. Music Player Screen Wireframe

- Music Library Screen

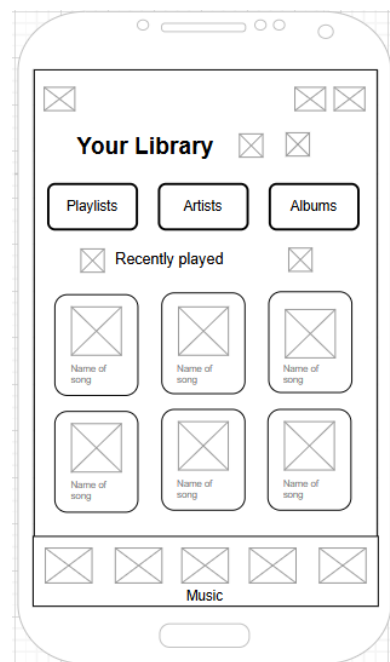


Figure 30. Music Library Screen Wireframe

- Music Player Screen

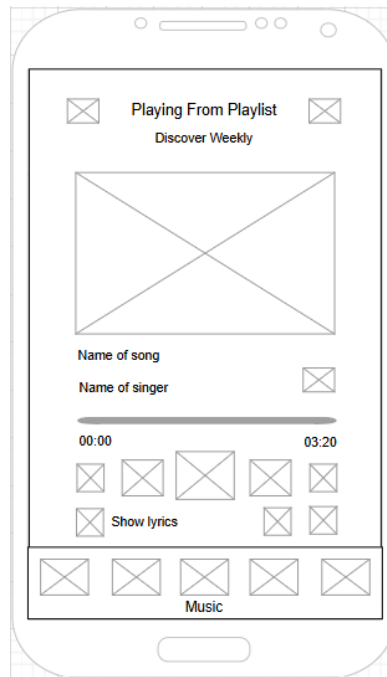


Figure 31. Music Player Screen Wireframe

- Show Lyrics Screen

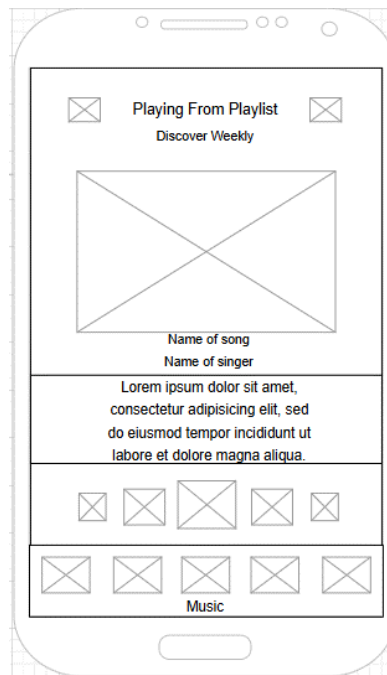


Figure 32. Show Lyrics Screen Wireframe

- Account Management Screen

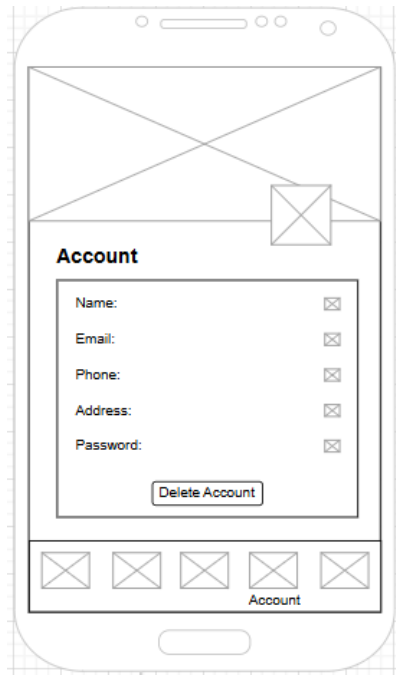


Figure 33. Account Management Screen

- Update Account Screen

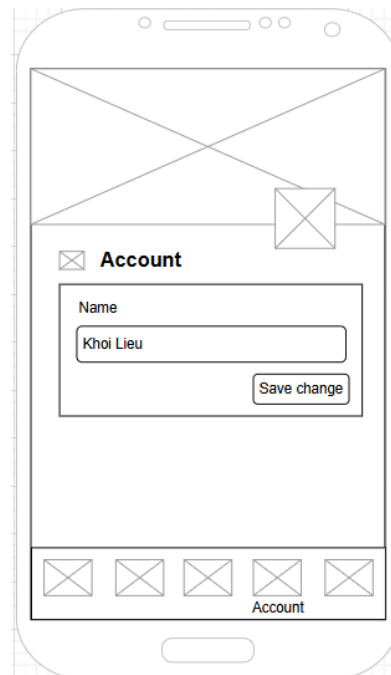


Figure 34. Update Account Screen Wireframe

- Setting Screen

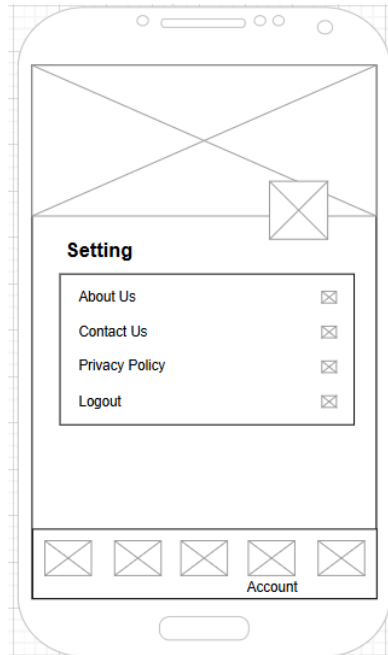


Figure 35. Setting Screen Wireframe

4.2 Mid-fidelity Prototype

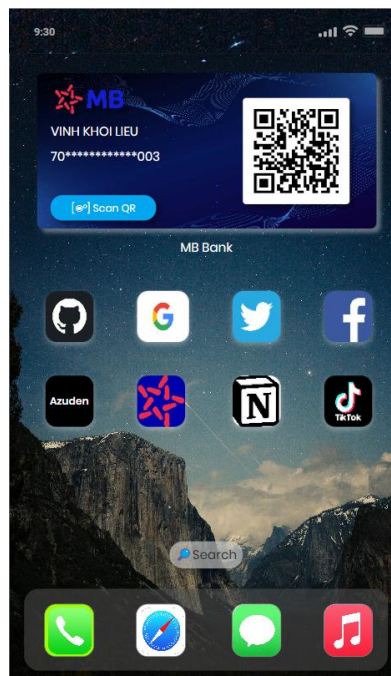


Figure 36. Azuden icon before accessing the app

Figure 36 shows user's phone screen before launching the app.

4.3 Welcome Screen

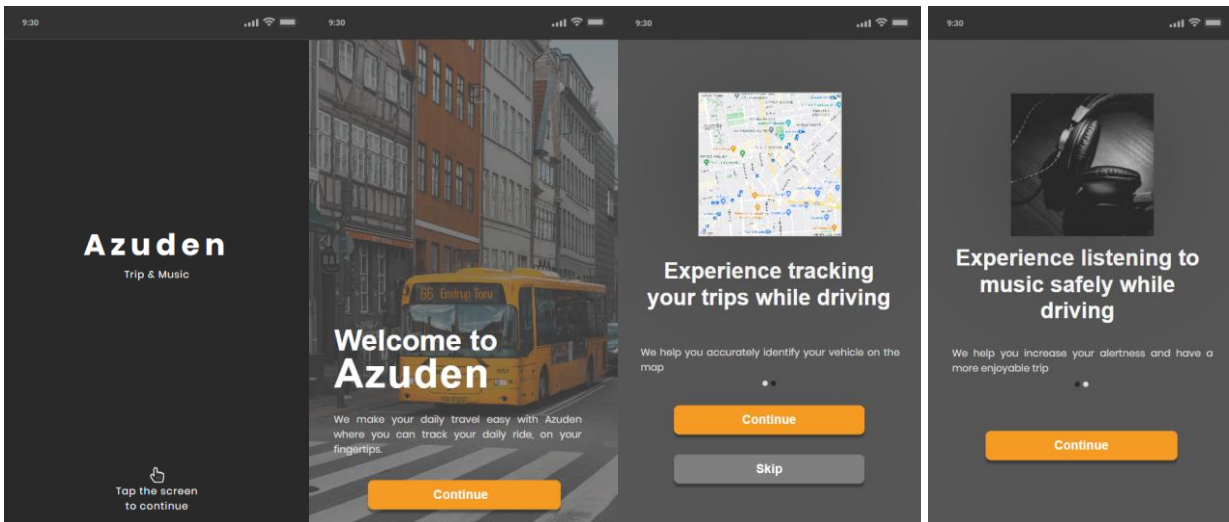


Figure 37. App welcome screens showcasing travel and music features.

Figure 37 showcases a series of prototype welcome screens for the Azuden application. The screens introduce users to the app's primary features, beginning with a minimalist initial screen featuring the Azuden logo and tagline "Trip & Music." Subsequent screens offer an overview of the app's functionality, including daily travel management, real-time trip tracking on a map, and enhancing safety through music playback while driving. The navigation buttons allow users to continue or skip the introductory guide, ensuring a user-friendly onboarding experience.

4.4 Sign-Up and Sign-In Screen

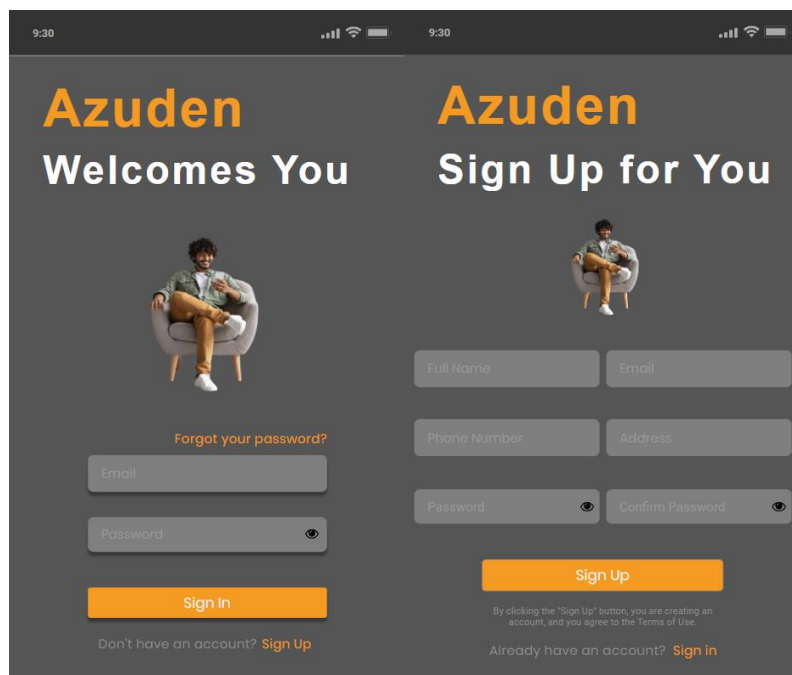


Figure 38. Azuden app's sign-in and sign-up screens for user authentication and registration.

Figure 38 presents the Azuden app's sign-in and sign-up screens. The sign-in page features input fields for email and password, along with options for password recovery and account registration. The sign-up screen allows users to register with fields for full name, email, phone number, address, and password confirmation. Both screens incorporate consistent branding with a central image and

welcoming text, offering a streamlined user experience for account access and creation.

4.5 Home Page Screen

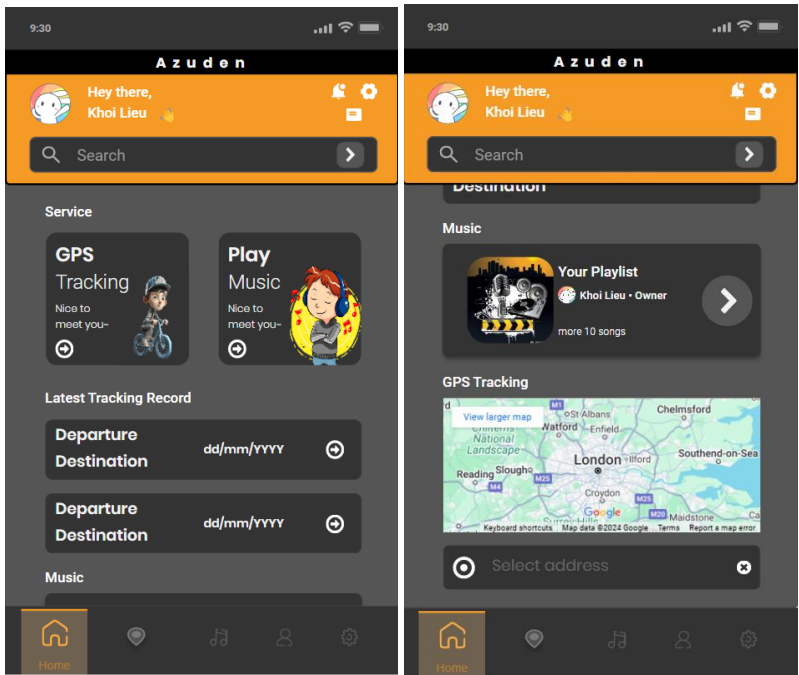


Figure 39. Azuden app's home page offers personalized GPS tracking, music services, and trip records.

Figure 39 displays the Azuden app's home page prototype, featuring a personalized greeting for the user, along with quick access to essential services such as GPS tracking and music playback. The upper section of the page provides a user profile, search bar, and settings icon, enhancing accessibility. Below, distinct cards offer GPS Tracking and Play Music options, each accompanied by a friendly introductory message. The middle section highlights recent tracking records, listing departure and destination details. Further down, users can access their customized music playlists and a GPS map with address input functionality for precise navigation. The interface is designed to streamline interaction and provide a comprehensive user experience.

4.6 Route Selection Screen

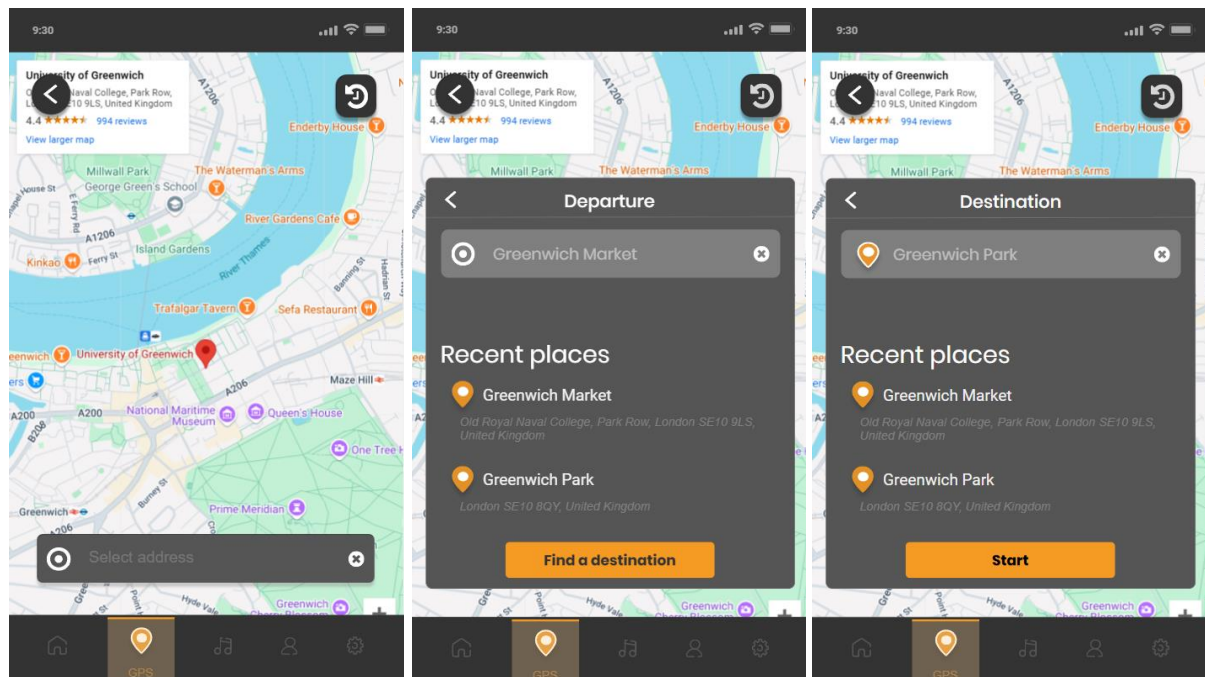


Figure 40. Route selection interface in the Azuden app for trip planning and tracking.

Figure 40 illustrates the route selection process for starting trip tracking within the Azuden app. The first screen presents a map interface for setting departure and destination points, displaying user location and relevant landmarks. Subsequent screens allow users to search for destinations, view recent places, and confirm route selections. This streamlined design emphasizes ease of navigation and route planning, facilitating a smooth start to trip tracking for motorcyclists.

4.7 Route Start Confirmation Screen

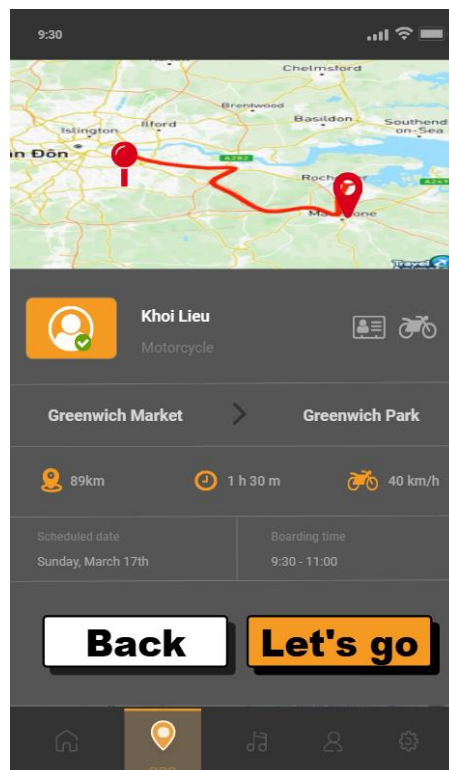


Figure 41. The trip confirmation screen displays route details, user info, and options to proceed or return.

Figure 23 presents the confirmation screen for trip details before starting journey tracking in the

Azuden app. It displays a route map, user profile, and essential trip data, including departure and destination points, estimated distance, travel time, and average speed. Additionally, it features a scheduled date and boarding time section. Two buttons, "Back" and "Let's go," allow users to either review or proceed with their planned trip, ensuring a comprehensive overview and control for an optimized travel experience.

4.8 Travel Route Tracking Screen

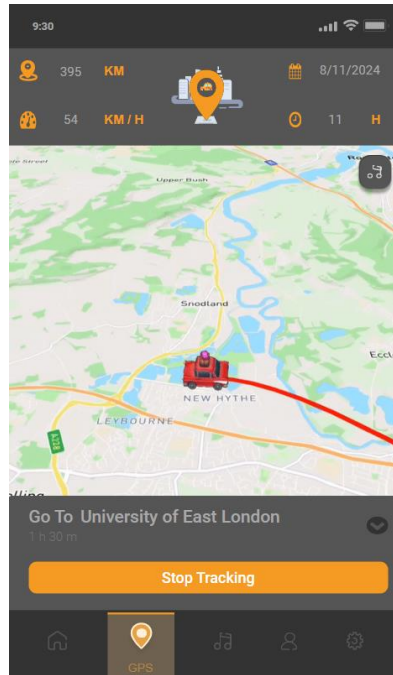


Figure 42. A live trip tracking screen displays route details and metrics, with a placeholder car icon used during development.

Figure 24 depicts the live trip tracking screen of the Azuden app, designed primarily for motorcyclists. The interface shows a route map, and key metrics like distance traveled, speed, duration, and an estimated completion time. While the display currently features a car icon to represent the user's position, this is due to development challenges in sourcing a suitable motorcycle icon for route simulation. The interface includes a "Stop Tracking" button for user control over their trip tracking experience.

4.9 Music Screen While Tracking The Screen

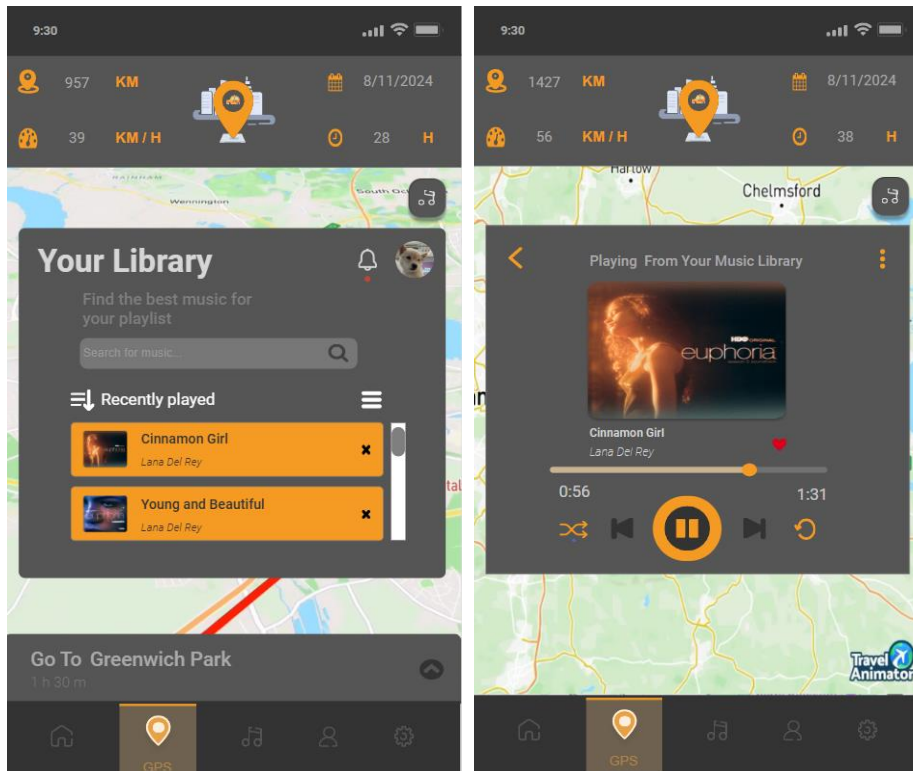


Figure 43. Trip tracking screen with a music player for seamless browsing and playback while traveling.

Figure 25 shows the trip tracking screen with an integrated music player in the Azuden app. Tapping the music icon brings up a library interface for browsing and selecting recently played songs. The playback screen offers controls for pausing, skipping, and volume adjustment, enhancing user convenience and safety by enabling music enjoyment during travel.

4.10 Travel Route Completion Screen

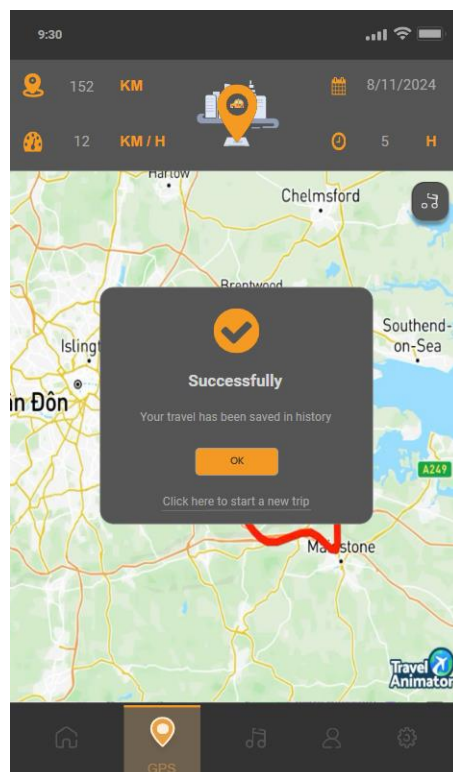


Figure 44. Trip completion notification confirming travel history save and offering options to continue or start a new trip.

Figure 26 illustrates the completion screen of the trip tracking feature in the Azuden app. Upon finishing a journey, a popup notification confirms that the travel record has been successfully saved in the user's history. The popup provides an "OK" button to acknowledge the message and an option to start a new trip, ensuring a smooth transition for continued use.

4.11 Tracking Record History Screen

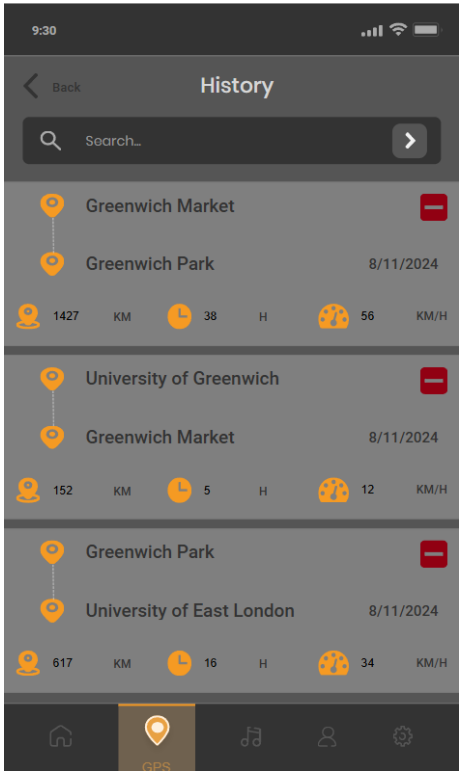


Figure 45. Trip history screen displaying completed journeys with details and options for search and record management.

Figure 27 shows the trip history screen in the Azuden app, displaying a list of completed journeys. Each entry includes key details such as departure and destination points, travel date, distance traveled, duration, and average speed. A search bar allows users to quickly find specific trips, while each history item features a delete button for record management. This interface ensures easy access and management of travel records for users.

4.12 Music Manager Screen Wireframe

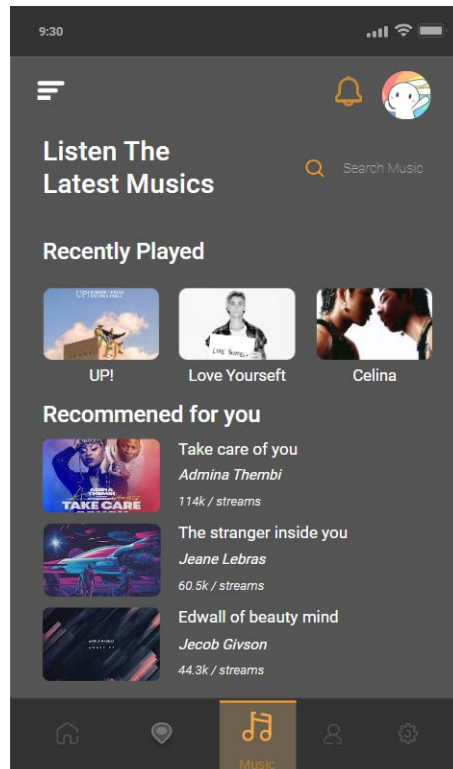


Figure 46. Music management screen with recently played and recommended tracks for personalized listening.

Figure 28 presents the Azuden app's music management screen, featuring a section for recently played tracks to give users quick access to their favorite songs. Below, tailored music recommendations are displayed, including play count details. A search bar allows for easy discovery of new music, enhancing user engagement through personalized listening and exploration options.

4.13 Music Library Screen

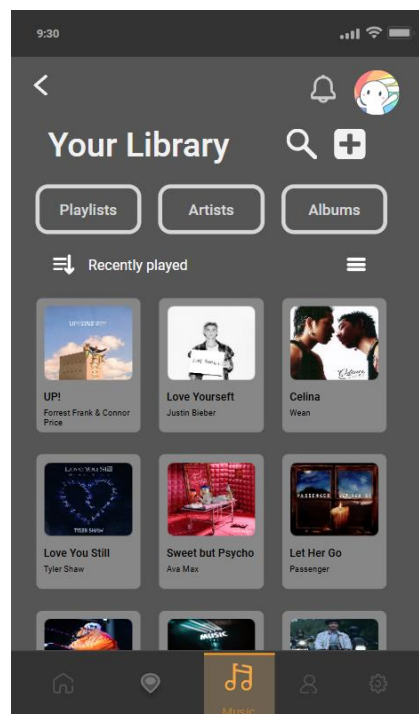


Figure 47. Music library with tabs for playlists, artists, albums, and recent tracks.

Figure 29 shows the music library screen in the Azuden app, allowing users to manage their music

collections. It includes tabs for browsing by playlists, artists, and albums, providing a tailored experience. Below, the "Recently played" section displays a grid of recently accessed songs for easy playback. The interface also features search and add options, enhancing the user's ability to organize and explore their music library efficiently.

4.14 Music Player Screen

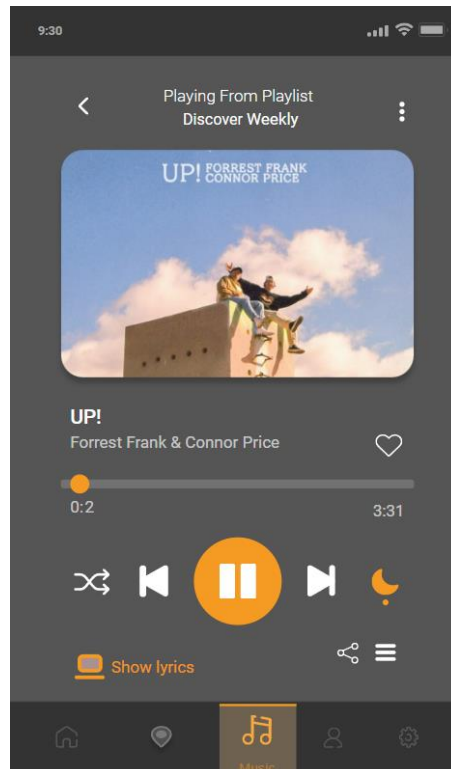


Figure 48. Music playback screen with controls for play, shuffle, repeat, and more.

Figure 30 displays the music playback screen in the Azuden app, featuring a song from a playlist with controls for play/pause, skip, shuffle, and repeat. Users can like the track, adjust playback, view lyrics, and access sharing options. This interface offers a seamless and engaging listening experience.

4.15 Show Lyrics Screen

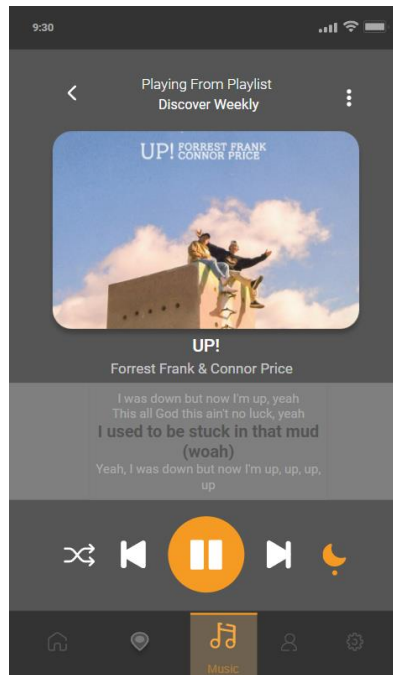


Figure 49. Lyrics display screen with real-time scrolling lyrics and music playback controls.

Figure 31 presents the lyrics display screen in the Azuden app's music player. While a song is playing, lyrics scroll beneath the playback controls, allowing users to follow along in real-time. The interface retains essential music controls, including play/pause, skip, shuffle, and repeat, ensuring a seamless listening and viewing experience.

4.16 Account Management Screen

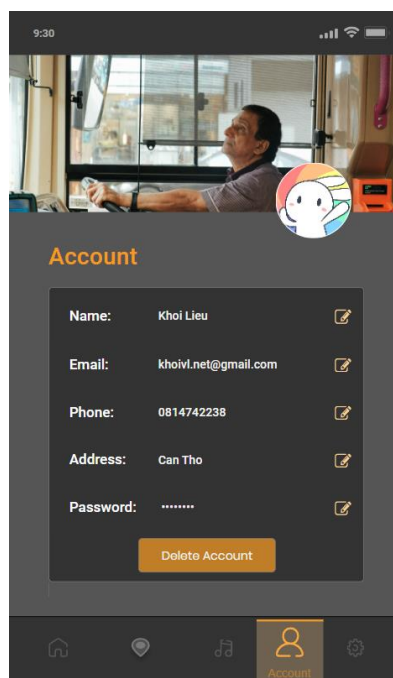


Figure 50. Account management screen with editable user profile details and a "Delete Account" option.

Figure 32 displays the account management screen in the Azuden app. It provides users with their profile details, including name, email, phone number, address, and password, all of which can be edited using the adjacent icons. A prominent "Delete Account" button is available for account removal. The interface ensures users have easy access to manage their personal information

securely.

4.17 Update Account Screen

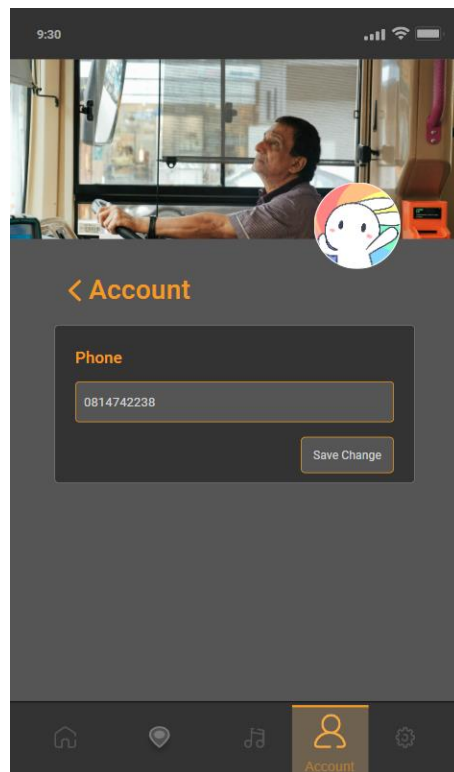


Figure 51. Edit phone number screen with a "Save Change" option.

Figure 33 shows the personal information update screen in the Azuden app. The interface allows users to edit their phone number with a simple input field and a "Save Change" button. The back arrow offers easy navigation to return to the previous account management screen, providing a straightforward and secure way to update personal details.

4.18 Setting Screen

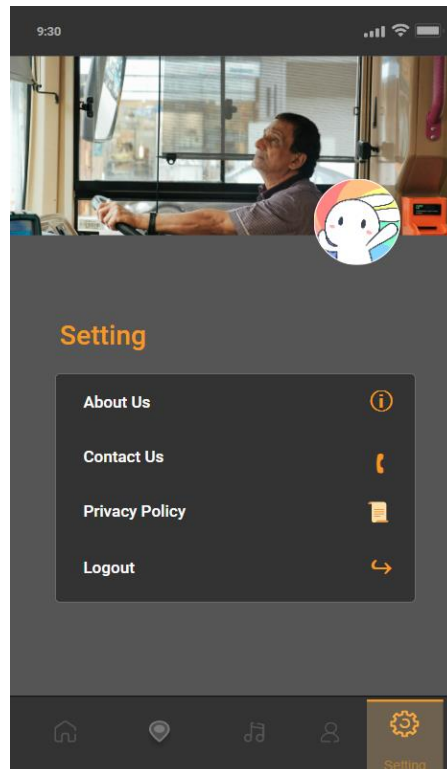


Figure 52. Settings screen with options for app info, contact, privacy, and logout.

Figure 34 displays the settings screen in the Azuden app, offering options such as "About Us," "Contact Us," "Privacy Policy," and "Logout." Each option is paired with a corresponding icon for clarity. This interface allows users to access app information, connect with support, review policies, or log out of their accounts easily.

5 Research Study

5.1 Research Objective

This study aims to evaluate the usability and effectiveness of the Azuden application prototype in a real-world environment. The specific objective is to confirm that the design of the application's components and interfaces are intuitive, user-friendly, and efficient, particularly in achieving the application's goal of helping users listen to music while tracking their motorbike travel routes. Special focus will be placed on ensuring minimal user distraction, seamless interaction with the interface, and maintaining rider safety during travel.

5.2 Hypotheses

- **Hypothese 1:** Users will experience the Azuden app interface easily and efficiently, allowing them to switch between music listening and route tracking functions without any hassle or excessive distraction while riding their motorbike.

Question:

- On a scale of 1 to 5 (1 being very difficult, 5 being very easy), how would you rate the ease of use of the Azuden app interface when switching between music playback and route tracking functions?
- Do you find the application's interface distracting while riding a motorcycle? (1 being not

distracting at all, 5 being very distracting).

- How would you rate the ease of finding and playing music within the app while tracking your route? (1 being very difficult, 5 being very easy).
- **Hypothesis 2:** The integration of music listening and route tracking functions on the Azuden app will enhance the user experience, create a sense of comfort and help them maintain motivation while traveling on a motorbike compared to using only the route tracking function alone.

Question:

- On a scale of 1 to 5, how satisfied are you with the GPS tracking feature of the Azuden app?
- Do you find the notifications and alerts of the app clear and helpful while traveling? (1 being not clear at all, 5 being very clear).
- On a scale of 1 to 5, how satisfied are you with the ability to adjust and customize settings within the app?
- How effective do you find Azuden in combining music playback and route tracking functions? (1 being not effective, 5 being very effective).
- Does the app meet your needs and provide comfort while riding your motorcycle? (1 being not comfortable, 5 being very comfortable).

5.3 Participants

The study will involve 10 to 12 participants, focusing on individuals who frequently utilize mapping services or other route-tracking applications. Participants will be recruited directly from social media platforms and specialized forums that host discussions related to navigation, commuting, and travel experiences.

5.4 Study Setup

Process: Participants will interact with the Azuden app prototype, performing tasks such as using the GPS tracking function to track their travel route, customizing and playing music during the trip, and then modifying app settings, editing account information to suit their needs.

Data Analysis: Feedback will be collected from participant responses using questionnaire to assess users' overall satisfaction, usability, and effectiveness with the app's features in real-world scenarios.

This is my survey link: <https://forms.gle/Vrtxk6NMKUHUagVz6>

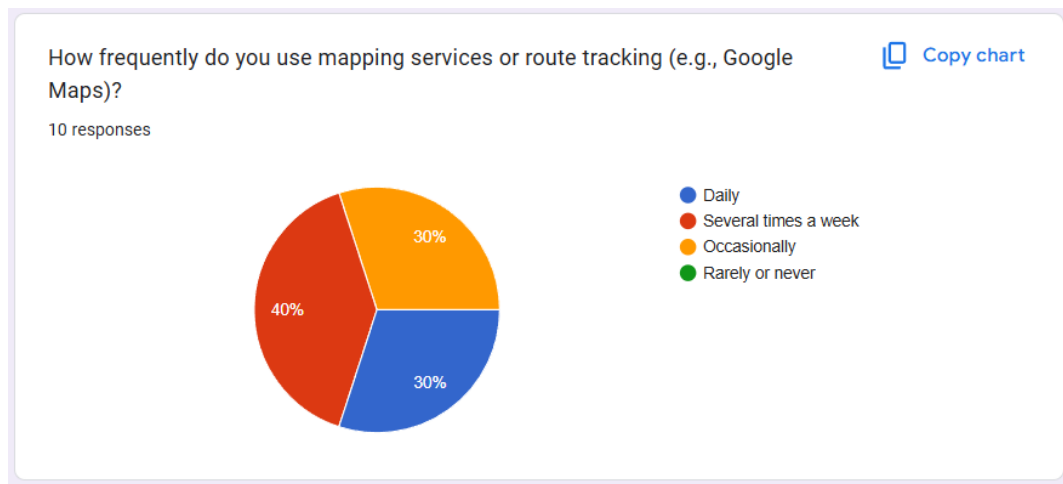


Figure 53. Data analysis for question 1

40% of users rely on mapping services daily, while 30% use them several times a week or occasionally, indicating frequent usage.

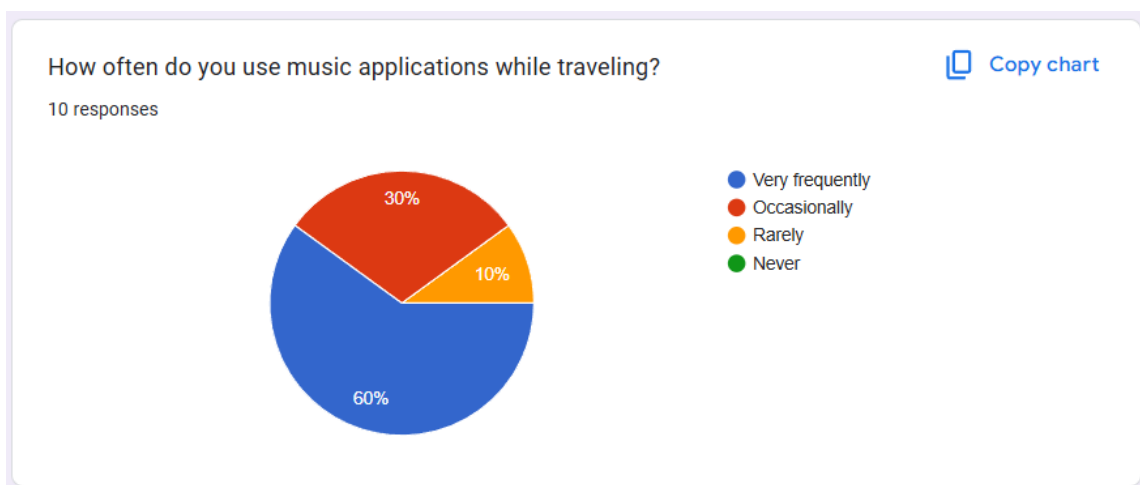


Figure 54. Data analysis for question 2

60% use music apps very frequently, showing music's importance during travel.

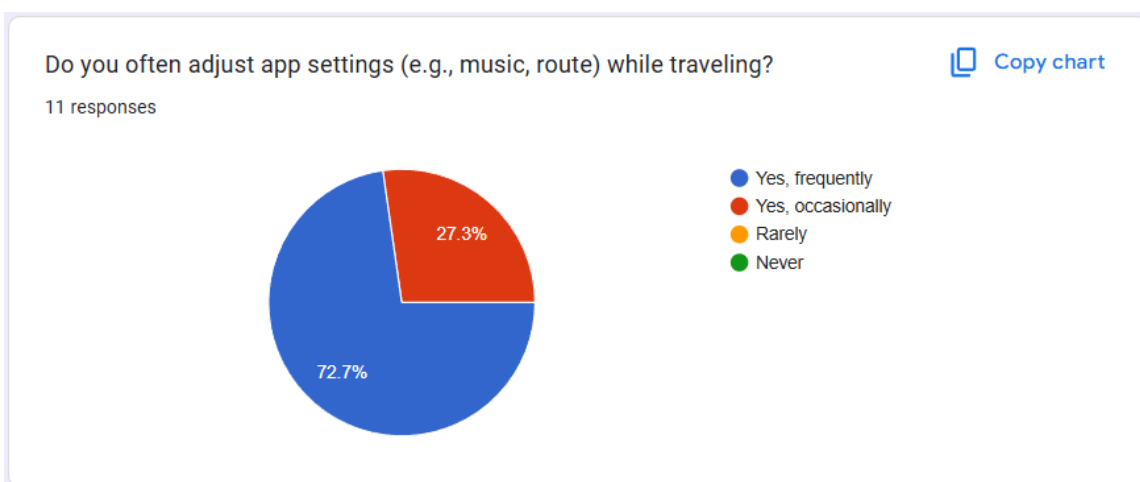


Figure 55. Data analysis for question 3

72.7% frequently adjust app settings, highlighting the need for quick access options.

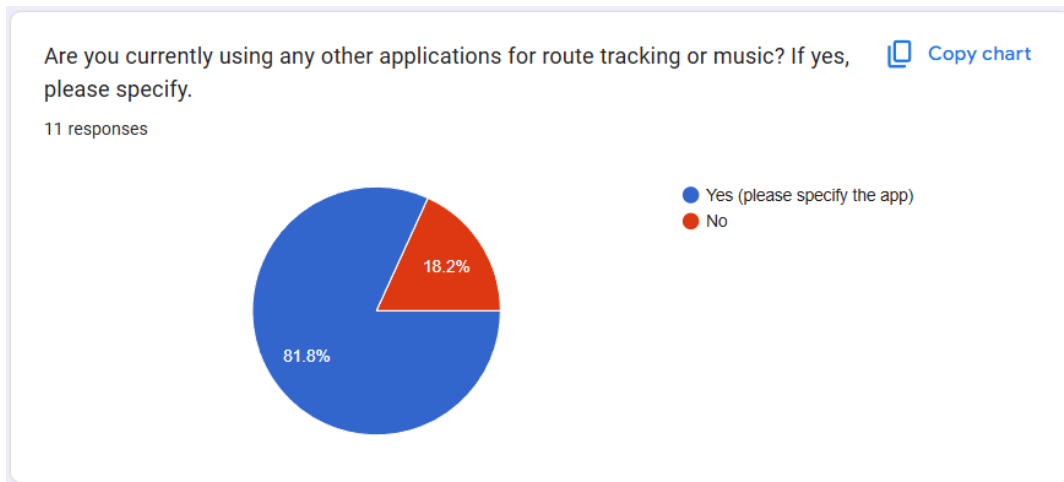


Figure 56. Data analysis for question 4

81.8% do not use other apps for tracking or music, suggesting Azuden could become a primary solution.

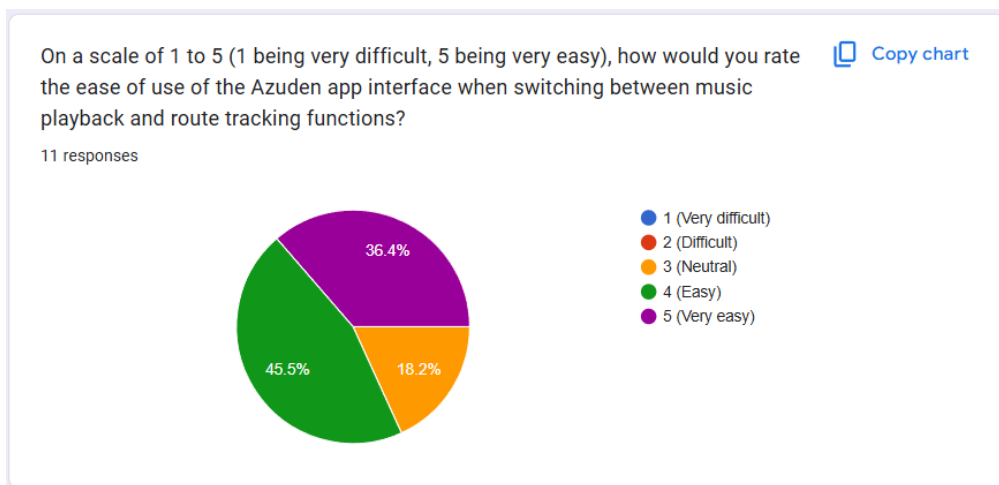


Figure 57. Data analysis for question 5

82% found it easy or very easy to switch between music and route tracking, showing a user-friendly interface.

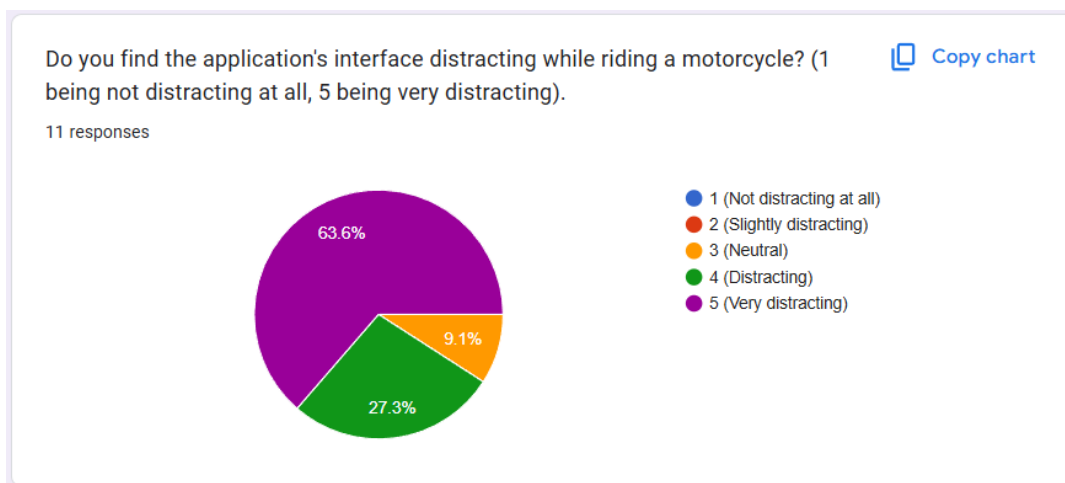


Figure 58. Data analysis for question 6

63.6% found the interface very distracting while riding, indicating a safety concern.

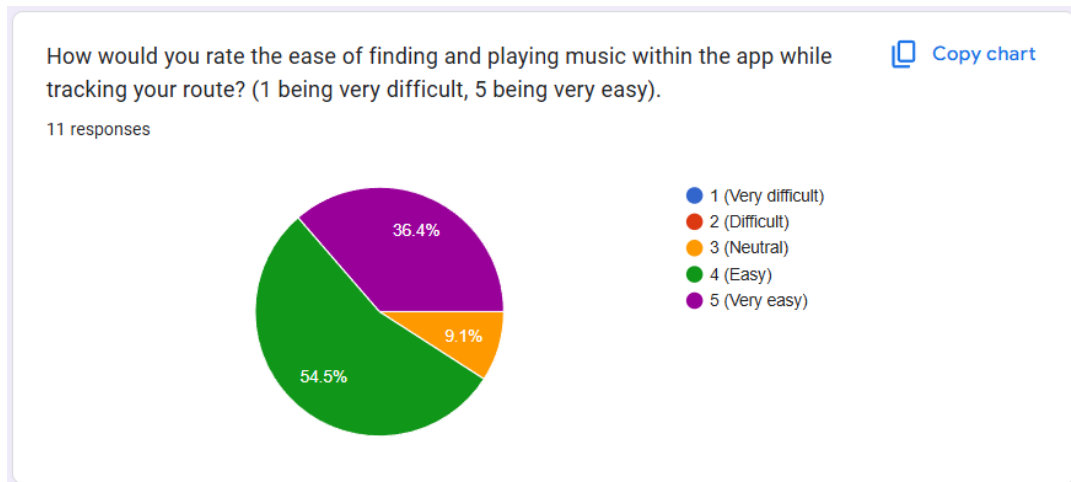


Figure 59. Data analysis for question 7

54.5% rated it very easy, with 36.4% finding it easy, indicating good accessibility.

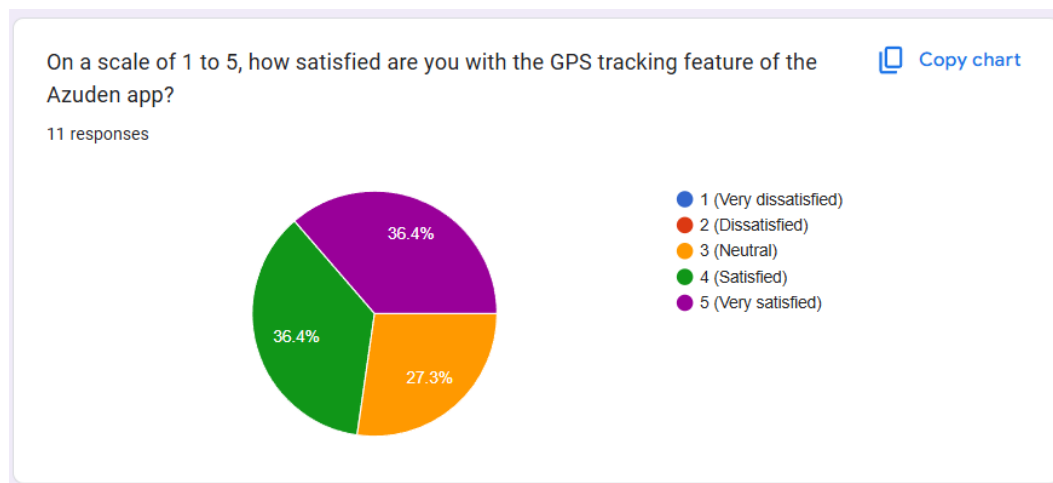


Figure 60. Data analysis for question 8

36.4% were very satisfied, with 27.3% satisfied, showing positive feedback overall.

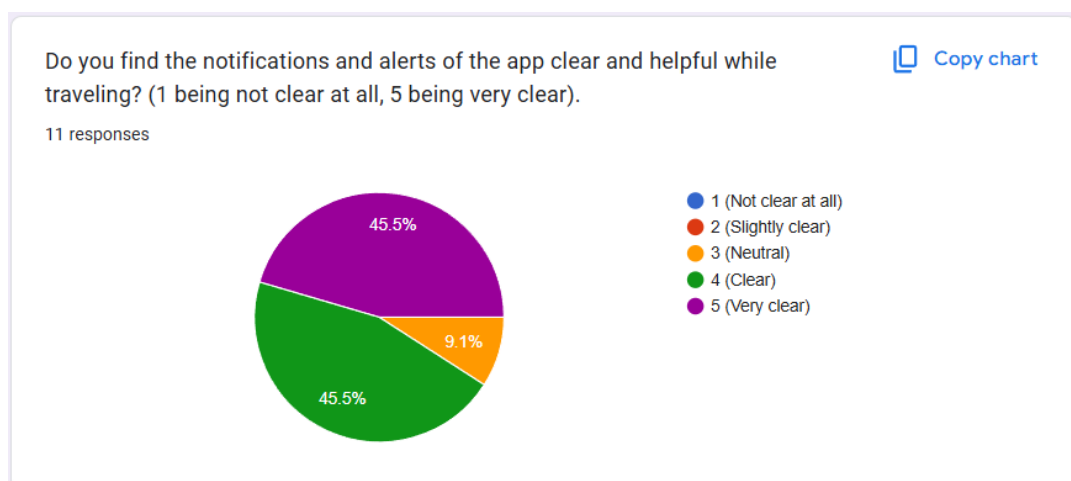


Figure 61. Data analysis for question 9

45.5% found notifications clear, and another 45.5% rated them very clear.

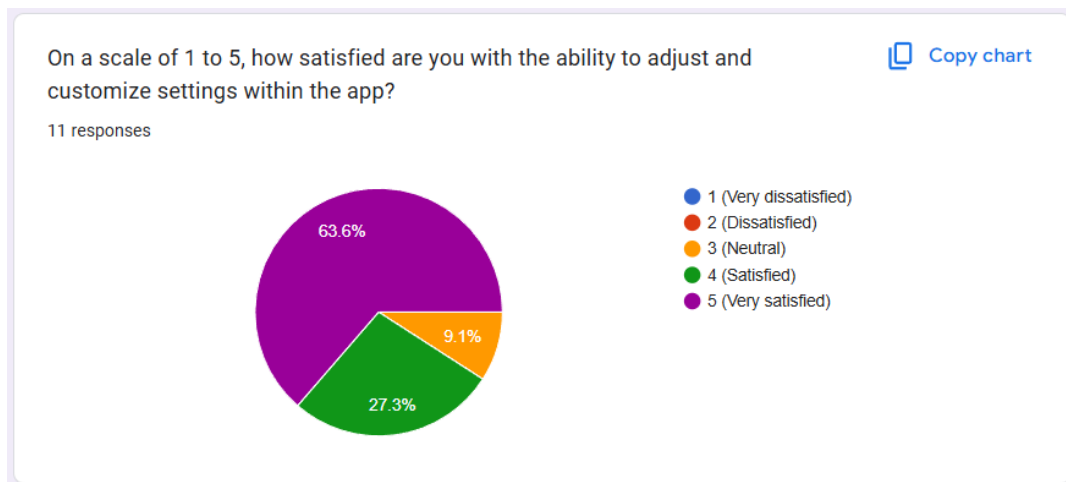


Figure 62. Data analysis for question 10

63.6% were very satisfied with customization features, highlighting strong user experience.

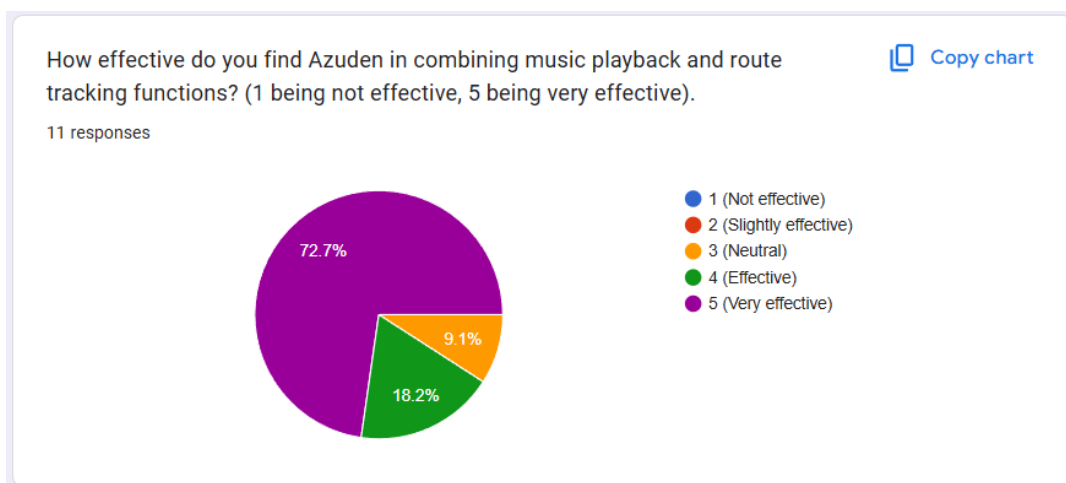


Figure 63. Data analysis for question 11

72.7% of respondents rated Azuden as "Very effective" in combining music playback and route tracking, with 18.2% finding it "Effective." Only 9.1% were "Neutral," indicating strong positive feedback overall.

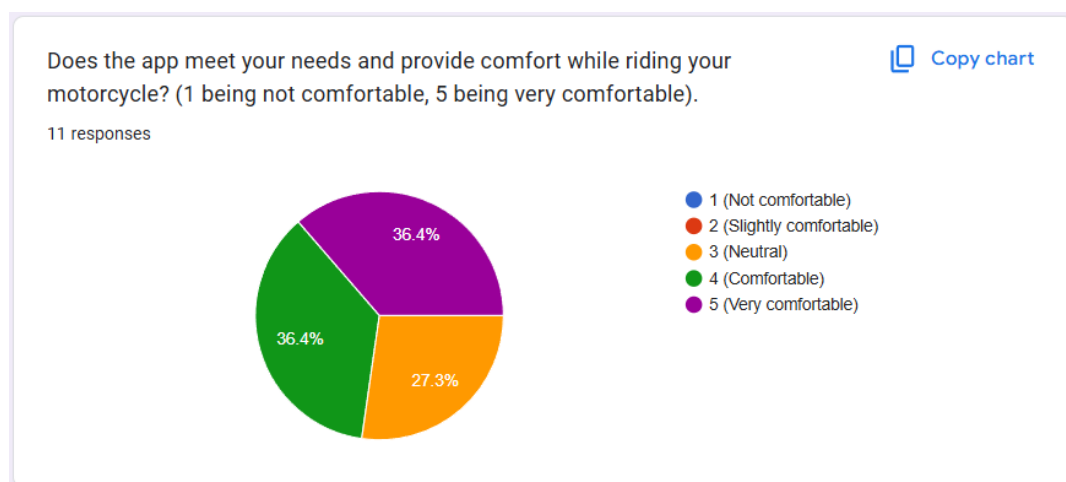


Figure 64. Data analysis for question 12

36.4% of respondents found the app "Very comfortable," and another 36.4% rated it "Comfortable"

while riding. 27.3% were "Neutral," suggesting overall satisfaction with some room for improvement.

6 Conclusion

The development of the Azuden application has provided valuable insights into enhancing the motorcyclist's experience by integrating music playback while GPS route tracking. However, several limitations and constraints emerged throughout the research and prototyping process. One significant challenge was ensuring a seamless interface design that minimized user distraction while maintaining safety during travel. Balancing functionality with simplicity in complex, real-time environments posed difficulties that required iterative testing and adjustments. Additionally, user recruitment presented constraints, as the sample size of participants was limited, potentially affecting the generalizability of feedback. Further refinement is necessary to accommodate diverse user needs, such as customizable safety alerts and voice commands to reduce interaction complexity. Potential next steps include expanding usability testing with a broader user demographic to validate design choices, incorporating AI-driven recommendations for music and routes, and refining data handling practices for enhanced security and privacy. Future iterations should continue to prioritize safety, user engagement, and the dynamic needs of motorcyclists in real-world conditions.

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Appendices

- Questions about user habits and service usage

Question 1:

How frequently do you use mapping services or route tracking (e.g., Google Maps)?

- ☐ Daily
- ☐ Several times a week
- ☐ Occasionally
- ☐ Rarely or never

This is a screenshot of a survey question. It features a light purple border and a white background. The question text is at the top, followed by four radio button options listed vertically.

Figure 65. Question 1

Question 2:

How often do you use music applications while traveling?

- ☐ Very frequently
- ☐ Occasionally
- ☐ Rarely
- ☐ Never

This is a screenshot of a survey question. It features a light purple border and a white background. The question text is at the top, followed by four radio button options listed vertically.

Figure 66. Question 2

Question 3:

Do you often adjust app settings (e.g., music, route) while traveling?

- ☐ Yes, frequently
- ☐ Yes, occasionally
- ☐ Rarely
- ☐ Never

This is a screenshot of a survey question. It features a light purple border and a white background. The question text is at the top, followed by four radio button options listed vertically.

Figure 67. Question 3

Question 4:

Are you currently using any other applications for route tracking or music? If yes, please specify.

- ☐ Yes (please specify the app)
- ☐ No

Figure 68. Question 4

- Quantitative questions for evaluating the Azuden application

Question 5:

On a scale of 1 to 5 (1 being very difficult, 5 being very easy), how would you rate the ease of use of the Azuden app interface when switching between music playback and route tracking functions?

- ☐ 1 (Very difficult)
- ☐ 2 (Difficult)
- ☐ 3 (Neutral)
- ☐ 4 (Easy)
- ☐ 5 (Very easy)

Figure 69. Question 5

Question 6:

Do you find the application's interface distracting while riding a motorcycle? (1 being not distracting at all, 5 being very distracting).

- ☐ 1 (Not distracting at all)
- ☐ 2 (Slightly distracting)
- ☐ 3 (Neutral)
- ☐ 4 (Distracting)
- ☐ 5 (Very distracting)

Figure 70. Question 6

- Question 7:

How would you rate the ease of finding and playing music within the app while tracking your route? (1 being very difficult, 5 being very easy).

☐ 1 (Very difficult)

☐ 2 (Difficult)

☐ 3 (Neutral)

☐ 4 (Easy)

☐ 5 (Very easy)

Figure 71. Question 8

- Question 8:

On a scale of 1 to 5, how satisfied are you with the GPS tracking feature of the Azuden app?

☐ 1 (Very dissatisfied)

☐ 2 (Dissatisfied)

☐ 3 (Neutral)

☐ 4 (Satisfied)

☐ 5 (Very satisfied)

Figure 72. Question 7

- Question 9:

Do you find the notifications and alerts of the app clear and helpful while traveling? (1 being not clear at all, 5 being very clear).

☐ 1 (Not clear at all)

☐ 2 (Slightly clear)

☐ 3 (Neutral)

☐ 4 (Clear)

☐ 5 (Very clear)

Figure 73. Question 9

- Question 10:

On a scale of 1 to 5, how satisfied are you with the ability to adjust and customize settings within the app?

- ☐ 1 (Very dissatisfied)
- ☐ 2 (Dissatisfied)
- ☐ 3 (Neutral)
- ☐ 4 (Satisfied)
- ☐ 5 (Very satisfied)

Figure 74. Question 10

- Question 11:

How effective do you find Azuden in combining music playback and route tracking functions? (1 being not effective, 5 being very effective).

- ☐ 1 (Not effective)
- ☐ 2 (Slightly effective)
- ☐ 3 (Neutral)
- ☐ 4 (Effective)
- ☐ 5 (Very effective)

Figure 75. Question 11

- Question 12:

Does the app meet your needs and provide comfort while riding your motorcycle? (1 being not comfortable, 5 being very comfortable).

- ☐ 1 (Not comfortable)
- ☐ 2 (Slightly comfortable)
- ☐ 3 (Neutral)
- ☐ 4 (Comfortable)
- ☐ 5 (Very comfortable)

Figure 76. Question 12